Rocket

Rocket wIntegrate

Installation and Configuration

Version 6.4.0

February 2016
WIN-640-ALL-IM-01

Notices

Edition

Publication date: February 2016
Book number: WIN-640-ALL-IM-01
Product version: Version 6.4.0

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Chapter 1: wIntegrate overview

wIntegrate is a connectivity tool that integrates host-based applications with the desktop. The initial interface is a terminal emulator with data transfer features designed for the multi-valued environments of UniVerse, UniData, D3, and mvBase. You can share data with popular Windows applications, including Microsoft Word and Excel.

For developers, a comprehensive programming language enables a range of options for integrating host environments with the desktop. You can add GUI features incrementally or create full GUI applications driven from the host.

A suite of host-based subroutines is supplied to control the PC. These subroutines are created for multi-valued environments, but the techniques used can be adapted to any host.

Developer tools include the Editor and a Tools menu with useful utilities. You can choose to display the Tools menu in **Setup > Preferences**. Select the **Scripts** tab and then use the **Menu Bar Script** drop-down list.

Introduction

This document provides a complete step-by-step guide to installing and setting up wIntegrate.

wIntegrate can be installed locally, as a thin client on Windows, or run as a thin client Java application/applet in a Web browser on Windows, Linux, and Macintosh. The Administrator and Monitor components are only applicable to the thin client installation.

wIntegrate features

All wIntegrate features are supported with Rocket's UniVerse, UniData, D3, and mvBase databases.

With wIntegrate, you can do the following tasks:

- Run concurrent sessions
- Customize a wIntegrate session
- Control any aspect of the application using powerful scripting language
- Optionally deploy a thin client on Windows or Java
- Run scripts from supplied host-based subroutines
- Define terminal emulations
- Export files from your PC to a host computer
- Import host computer files to your PC
- Transfer files from one host computer to another host computer
- Simplify deployment through optional thin client and browser interfaces

Chapter 2: Installing wIntegrate

There are a few ways to install wIntegrate, locally being the preferred method.

Local client installations require less administrative overhead as well as no need for separate wIntegrate Server hardware and infrastructure. The program communicates directly with the host, so local installations run faster than other installation methods. For more information about the local installation, see Installing the wIntegrate client locally, on page 9.

Thin client installations are used with administration in mind and for larger installation sites. You can install the wIntegrate Server, and then install a Windows or Java version of the thin client. For more information about the thin client installation, see Installing the wIntegrate Server, on page 11.

Both the local and thin clients allow for easy upgrading. For more information, see Upgrading wIntegrate, on page 16.

wIntegrate also supports globalization. For more information, see Globalization, on page 46.

System requirements

The following system requirements are necessary for both the local client and the server:

wIntegrate local client requirements

- Windows 8 Professional, 32 and 64-bit
- Windows 8.1 Professional, 32 and 64-bit
- Windows 7 Ultimate SP1, 32 and 64-bit
- Windows Server 2012
- Windows Server 2008 SP2
- Processor compatible with the operating system
- 30 MB available hard disk space for complete installation of program and support files
- TCP/IP-compatible network card (100 Mbits or better), or serial port

wIntegrate Server requirements

- Windows 8 Professional, 32 and 64-bit
- Windows 8.1 Professional, 32 and 64-bit
- Windows 7 Ultimate SP1, 32 and 64-bit
- Windows Server 2012
- Windows Server 2008 SP2
- 2.0 GHz processor or better, compatible with the operating system
- 600K memory per connected user
- 20 MB available hard disk space for complete installation of program and support files, plus 1 MB per user for session files
- TCP/IP-compatible network card
- 1.0 Gbps network

wIntegrate Windows thin client requirements

- Windows 8 Professional, 32 and 64-bit
- Windows 8.1 Professional, 32 and 64-bit
- Windows 7 Ultimate SP1, 32 and 64-bit
- Windows Server 2012
- Windows Server 2008 SP2
- Processor compatible with the operating system
- 10 MB available hard disk space for complete installation of program and support files
- TCP/IP-compatible network card
- 1.0 Gbps network

wIntegrate Java thin client requirements

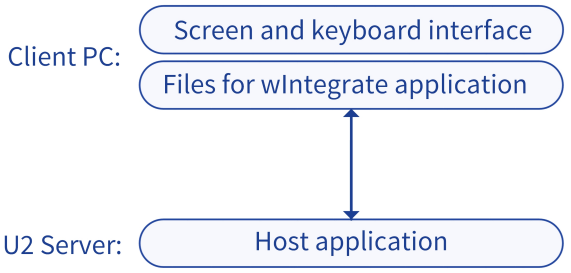
- Java 7 or later
- 32-bit and 64-bit versions of Windows 8 and Windows 7 with Internet Explorer 9, 10, 11, Firefox 22-30, and Google Chrome 30-35 (all supported combinations)
- Red Hat Enterprise Linux 5, 6, and 7 with Firefox 30
- SuSE Linux Enterprise Desktop 10 SP2 with Firefox 30
- Mac OS X 10.8 (Mountain Lion), 10.9 (Mavericks), 10.10 (Yosemite) with Safari

wIntegrate client local installation

wIntegrate is installed locally, built from ActiveX components.

The interface is integrated with session ActiveX control communicating with the host application, or U2 Server, as shown in the following figure.

Figure 1: Local installation



The Windows Installer is a standard Windows application that handles the installation, maintenance, and uninstallation of applications on recent Windows platforms. It requires that applications to be installed are stored in a standard `.MSI` file structure. `.MSI` files are available to developers who want to create custom installers. They are included on the product CD, though not referenced on the menus. Five installation executables are available for the product, described in the following table.

Table 1: Installation executables

Executable	.MSI file	Installs component
`Setup.exe`	`wIntegrate.msi`	Classic local installation version

Executable	.MSI file	Installs component
`SetupServer.exe`	`wIntegrate_Server.msi`	wIntegrate Server
`SetupClient.exe`	`wIntegrate_client.msi`	Windows thin client
`SetupAdmin.exe`	`wIntegrate_Administrator.msi`	wIntegrate Administrator
`SetupMonitor.exe`	`wIntegrate_Monitor.msi`	wIntegrate Monitor

Installing the wIntegrate client locally

wIntegrate is installed on a local computer to the `C:\Program Files (x86)\wIntegrate` location using an installation wizard.

Prerequisites

- Review the System requirements, on page 7.
- You must have a serial number or key to enter for the installation; otherwise, you are only able to use the 10-day single user evaluation. If you do not have a serial number or key, you can contact u2bc@rocketsoftware.com to obtain one.
- Close all other open applications before beginning the installation.
- Optionally create an install log, as described in Creating an install log, on page 16.

Procedure

1. From the top level of the product setup image, run the `AutoRun.exe` file.
2. From the wIntegrate menu, click **wIntegrate - local install**.
3. Follow the steps in the installation wizard.
   a. On the Setup Type page, select **Typical** or **Custom** installation.

      If you choose **Custom**, there are five options available. wIntegrate Application, Host Programs, and Examples are enabled by default. You can also choose to install Developer Additions and Network Administrator Tools. For more information about the developer additions, see the *wIntegrate User Guide*.

   b. Click **Install**.

      A licensing dialog box appears.
4. In the wIntegrate Licensing dialog box, enter your user name and organization. Enter the serial number and key that came with your software package. Click **OK**.

   If you do not have a license, you can select the **10-day single user evaluation** option.

   > **Note:** The **Serial Number** and **Key** fields are not case-sensitive, and the dashes in the key number are for legibility only; you do not need to type the dashes.

5. When wIntegrate is finished installing, click **Finish**.

Results

When you start wIntegrate for the first time, the program checks to see whether there is a `wIntegrate` folder in `C:\Users\`*`myname`*`\Documents\wIntegrate`. If not, the `wIntegrate` folder is created, relevant Registry settings are created, and the following files are copied from `C:\Program Files (x86)\wIntegrate`:

- Session files

- Color settings
- Default settings
- File transfer parameters
- Saved queries
- Saved templates
- Schedule parameters

Evaluating a newer version

The full license for your installed version works for that version and earlier versions. It does not work for later versions. If, for example, you have installed v6.3.6 with a full license, and want to evaluate v6.3.7, you must remove the v6.3.6 license temporarily before creating the evaluation license for v6.3.7. After the evaluation, reinstate the full license for v6.3.6.

1. With a full license for v6.3.6 installed, start Windows Explorer and go to folder `C:\ProgramData\Rocket Software\wIntegrate License`. Locate the `license.wil` file.
2. The `license.wil` file contains your current license information. Rename the file from `license.wil` to, for example, `license.wil636`. Note the license file name before changing it.

   You can delete this file. However, if you do, be sure to have your original Serial Number and License Key available for when you reinstall the current version later in the process.
3. Install wIntegrate v6.3.7 (or the version you want to evaluate), making sure an earlier version is not running. You do not need to uninstall the earlier version first, as it will be upgraded. You do not need to backup your settings in `My Documents\wIntegrate` as these are generally compatible across different versions.

   If you are prompted for a license, select **10 day single user evaluation**, and click **OK**. A new `license.wil` file is written to `C:\ProgramData\Rocket Software\wIntegrate License`.

   > **Note:** You should avoid installing the new version to a different folder, resulting in two different versions installed on the same machine. Some components are referenced by entries in the Registry and conflicts can cause unexpected errors. You can see any conflicts by running **Check Installation** from the **Start > wIntegrate** menu.

4. When you start evaluating the new version, use **File > Save As** to save the session with a new name for the evaluation.

   When you save the session, there may be new session parameters in the saved session `.WIC` configuration file. Generally, if you open this new session `.WIC` file in an older version of the program, any unrecognized session parameters are discarded. However, if you use a new value for an existing parameter, the old version of the wIntegrate will error. For example, if you are running v6.2.0 and install an evaluation of v6.3.7 to test the TLSv1.2 protocol in SSL communications and you save the configuration `.WIC` file, when you reinstall v6.2.0 wIntegrate will error because the TLSv1.2 protocol is not known in v6.2.0.

   > **Note:** If you installed wIntegrate with the Developer Additions enabled, you can use the Editor to edit session `.WIC` files and change values manually.

5. When you have completed the evaluation, uninstall the newer version and delete the evaluation `license.wil` file from `C:\ProgramData\Rocket Software\wIntegrate License`.

Uninstalling does not affect your settings in `My Documents\wIntegrate`.

6. If you renamed the `license.wil` file in step 2 above, rename it to its original name.
7. Reinstall your older version. Any existing license file will be shown in the licensing dialog. If you left the newer evaluation license file in place, the wIntegrate determines the evaluation is still in progress, but for the earlier version.

wIntegrate thin client installation

In larger installations, the thin client reduces cost of ownership and simplifies administration. The application is split into two separate parts: the screen and the keyboard interface, and the underlying functionality. These parts are respectively the Windows thin client and the wIntegrate Server. Both parts can run on the same computer. A single wIntegrate Server can support multiple Windows and Java thin clients. The thin client can run on the same computer as the wIntegrate Server.

Definitions for dialog boxes, menus, and so on are stored with the wIntegrate Server and transferred to the desktop immediately before being displayed.

A common code base is used for the Windows thin client and local client, so any change made for one version is automatically made for the other. This reduces the development overhead and maximizes compatibility.

The Windows thin client is almost indistinguishable from the local install version. The Java client looks different because it is rendered by Java instead of Windows, but the functionality is almost identical.

Installing the wIntegrate Server

The wIntegrate Server is normally installed to a central server for access by multiple clients. The wIntegrate Server does not distinguish between the Windows and Java thin clients. Though it knows which one is connected on each session, it sends and receives data to each in the same way.

Prerequisites

- Review the System requirements, on page 7.
- You must have a serial number or key to enter for the installation; otherwise, you are only able to use the 10-day single user evaluation. If you do not have a serial number or key, you can contact u2bc@rocketsoftware.com to obtain one.
- Close all other open applications before beginning the installation.
- Optionally create an install log, as described in Creating an install log, on page 16.

About this task

The wIntegrate Server runs as a service. This means it runs automatically as a background process and has no native interface—the wIntegrate Administrator and wIntegrate Monitor act as the interface. The wIntegrate Server can be controlled like any other service by using Windows utilities or from the wIntegrate **Server** menu using the Service Control Program. Note that the Service Control Program must be started with **Run as administrator**, otherwise it may not have the necessary privileges to access system services.

You can put the wIntegrate Server on the same machine as your database, which will save one stage of the communications.

Procedure

1. From the top level of the product CD setup image, run the `AutoRun.exe` file.

   On Windows 7, Windows 8, Windows Server 2008, or Windows Server 2012, you must right-click `AutoRun.exe` and select **Run as administrator**, otherwise the installer may not have the necessary privileges to access system services.
2. From the wIntegrate menu, click **wIntegrate - thin client install menu**.
3. Click **Install wIntegrate Server**.
4. Follow the steps in the installation wizard.
   a. On the Setup Type page, select **Complete** or **Custom** installation.

      If you choose **Custom**, there are four options available. wIntegrate Server Application, Host Programs, Examples, and Java Client Component are enabled by default. For more information about the Java thin client, see Java thin client installation, on page 15.

   b. Click **Install**.

      A licensing dialog box appears.
5. In the wIntegrate Licensing dialog box, enter your user name and organization. Enter the serial number and key that came with your software package. Click **OK**.

   If you have installed and licensed the local client, the fields are automatically populated with the same information. If you do not have a serial number or key, you can contact u2bc@rocketsoftware.com to obtain one; otherwise you can select the **10-day single user evaluation** option.

   Note: The **Serial Number** and **Key** fields are not case-sensitive, and the dashes in the key number are for legibility only; you do not need to type the dashes.

6. When the wIntegrate Server is finished installing, click **Finish**.

Results

For thin client users, all of the user settings are stored on the server. When a thin client user connects to the wIntegrate Server for the first time, the program checks whether the thin client login name is in the folder `C:\ProgramData\Rocket Software\wIntegrate Server\Users`. If not, a folder is created on the server with the thin client user name, and the following files are copied from `C:\Program Files (x86)\wIntServ`:

- Session files
- Color settings
- Default settings
- File transfer parameters
- Saved queries
- Saved templates

Installing the Administrator

The Administrator is the interface for the wIntegrate Server. It shows the currently connected thin client users and lets you set options. It is automatically installed with the wIntegrate Server. You can also install it separately to any computer and run it remotely, subject to remote access being enabled.

Prerequisites

- Review the System requirements, on page 7.
- Close all other open applications before beginning the installation.
- Optionally create an install log, as described in Creating an install log, on page 16.

About this task

Authorized users can run the Administrator on their own computer that is connected to a wIntegrate Server that is running on a different computer, provided that the user has authority to run the Administrator connected to this wIntegrate Server, and that the Enable Remote Administration option is active. For more information, see wIntegrate Administrator, on page 27.

Procedure

1. From the top level of the product CD setup image, run the `AutoRun.exe` file.

   On Windows 7, Windows 8, Windows Server 2008, or Windows Server 2012, you must right-click `AutoRun.exe` and select **Run as administrator**, otherwise the installer may not have the necessary privileges to access system services.
2. From the wIntegrate menu, click **wIntegrate - thin client install menu**.
3. Click **Install wIntegrate Administrator**.
4. Follow the steps in the installation wizard.
5. When the wIntegrate Administrator is finished installing, click **Finish**.

Installing the Monitor

The wIntegrate Monitor is an extended version of the Administrator. It allows authorized users to view user sessions on the wIntegrate Server in real time. This is useful for support. The Monitor must be installed separately because it is not installed with the Server.

Prerequisites

- Review the System requirements, on page 7.
- Close all other open applications before beginning the installation. This enables any necessary system DLLs to be installed to your computer without having to restart.
- Optionally create an install log, as described in Creating an install log, on page 16.

Procedure

1. From the top level of the product CD setup image, run the `AutoRun.exe` file.

   On Windows 7, Windows 8, Windows Server 2008, or Windows Server 2012, you must right-click `AutoRun.exe` and select **Run as administrator**, otherwise the installer may not have the necessary privileges to access system services.
2. From the wIntegrate menu, click **wIntegrate - thin client install menu**.
3. Click **Install wIntegrate Monitor**.
4. Follow the steps in the installation wizard.
5. When the wIntegrate Monitor is finished installing, click **Finish**.

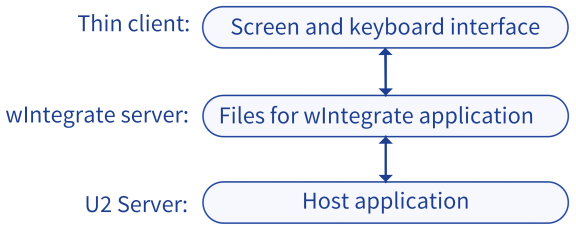
Windows thin client installation

The Windows thin client relies on the wIntegrate Server on a separate computer. The thin client is a Windows application, and indistinguishable in use from the local installation, but it has to be installed on the client PC.

The Windows thin client looks and feels like the regular local installation, so there are no backwards compatibility or restraining issues.

Unlike the local installation, the thin client normally stores user files on the wIntegrate Server. These session files and other custom settings are available from any connected client and loaded when the user logs in to the wIntegrate Server. So the user's settings are not tied to a specific client device, but are loaded regardless of the client device used.

Figure 2: Windows 3-tier thin client installation



Installing the Windows thin client

Prerequisites

- Review the System requirements, on page 7.
- Close all other open applications before beginning the installation.
- Optionally create an install log, as described in Creating an install log, on page 16.

Procedure

1. From the top level of the product CD setup image, run the `AutoRun.exe` file.

   On Windows 7, Windows 8, Windows Server 2008, or Windows Server 2012, you must right-click `AutoRun.exe` and select **Run as administrator**, otherwise the installer may not have the necessary privileges to access system services.
2. From the wIntegrate menu, click **wIntegrate - thin client install menu**.
3. Click **Install wIntegrate thin client (Windows)**.
4. Follow the steps in the installation wizard.
5. When wIntegrate is finished installing, click **Finish**.

Results

For thin client users, all of the user settings are stored on the server. When a thin client user connects to the wIntegrate Server for the first time, the program checks whether the thin client login name is in the folder `C:\ProgramData\Rocket Software\wIntegrate Server\Users`. If not, a folder is created on the server with the thin client user name, and the following files are copied from `C:\Program Files (x86)\wIntServ`:

- Session files
- Color settings

- Default settings
- File transfer parameters
- Saved queries
- Saved templates

Starting the Windows thin client

After the Windows thin client is installed, you can begin using it.

1. Click the Windows **Start** button and select **All Programs > wIntegrate Client > wIntegrate Client**.
2. Enter your user name, password, and the name of the server that you want to log in to.

   The user name, password, and privileges are obtained from either Windows Active Directory, or from the details set up in User Management.

   The host screen appears.

Java thin client installation

The Java thin client relies on the wIntegrate Server running as a Windows application on a separate computer. Its functionality is almost indistinguishable from the local installation. It runs in the Java Runtime Environment on Windows, Macintosh, and Linux.

The Java thin client is installed during the complete installation of the wIntegrate Server. See Installing the wIntegrate Server, on page 11 for more information.

The Java thin client is not completely backwards compatible because some low-level Windows functionality is not available. See Missing features in the Java client, on page 42 for more information.

The Java thin client is supplied as a `.JAR` file that can run in two ways. First, as an application with the same overall look and feel as the local version and the Windows thin client. Second, it can run as a Java applet in a browser. This means that, unlike the Windows thin client, there is no need for prior installation because the applet is downloaded at run time. (See the sample HTML files in `C:\Program Files\wIntServ\wIntJava`.)

The default location for files is the `C:\ProgramData\Rocket Software\wIntegrate Server\Users\`*`myname`* location, where *myname* is the user name that is entered when logging on to the wIntegrate Server. A user can connect from any supported web-based device and use the same settings.

The Java thin client can be deployed for in-house users or on the web. When running on the web, connect through the firewall on the web server, using a small redirector applet as a bridge between the client and Server. When being used in-house, there is no need for the redirector applet.

Installing the Java thin client

The Java thin client is installed during the complete installation of the wIntegrate Server.

See Installing the wIntegrate Server, on page 11 for more information about installing the Java client.

Starting the Java thin client

After the Java thin client is installed, you can begin using it.

Prerequisites

The Java 7 (or later) Runtime Environment must be installed on your desktop.

Important: Problems can occur while using the Java thin client if you have multiple versions of Java installed on your machine. It is recommended that you uninstall older versions of Java before installing and running the Java thin client.

Procedure

1. Click the Windows **Start** button and select **All Programs > wIntegrate Server > wIntegrate Java Client**.
2. Enter your user name, password, and the name of the server that you want to log in to.

   The host screen appears.

Creating an install log

You can create a log showing what files were installed and how the Registry was updated.

For example, you can create a log for `Setup.exe` in the `C:\Temp\wint_install.log` file by using the following command line:

```
Setup.exe /v"/l*v \"C:\Temp\wint_install.log\""
```

The `/v` command causes parameters to be passed to the embedded `.MSI` files. For more command line options, see the information on the InstallShield web site.

Upgrading wIntegrate

The upgrading process is similar to the installation process. You can upgrade from v4.2.0 and above.

We recommend upgrading to the latest version whenever possible to benefit from new features and stability improvements. This is particularly true of the thin client deployment where stability of the wIntegrate Server is critical.

Upgrading the host programs

When you upgrade either the local installation or the wIntegrate Server, we recommend that you install the latest version of the host programs.

We occasionally introduce new features which require later host programs. Also, we sometimes improve stability in existing host programs, but ensure that they are 100% backwards-compatible with earlier releases.

When we want to enhance a program/subroutine in such a way that it would break backward-compatibility, a new subroutine appended with a number is created. For example, compare the subroutines WIN.PCRUN (two parameters) and WIN.PCRUN2 (four parameters). We recommend that you use the latest available version.

The host program version is stored in the item HOSTVER in the WIN.PROGS file. When you run Query Builder, the version is shown along with the host MACHINE.TYPE in the blue box on the emulation screen.

See Installing host programs, on page 21 and Sharing the host programs, on page 23.

Upgrading the client from an earlier version

When you are upgrading from an earlier version, such as v4.2, where user settings were stored in Program Files, the user files are preserved and copied automatically to the `C:\Users\myname\Documents\wIntegrate` folder. This folder is the default location for files that are saved by processes such as importing data, editing records, and more.

Note: If you have already been using wIntegrate, make a separate copy of any files in the Program Files folder that you have modified for your own use.

The upgrading procedures are the same as described in Installing the wIntegrate client locally, on page 9.

Administrators can customize the installation process by editing the `winteg.wip` file in `C:\Program Files (x86)\wIntegrate`.

The `winteg.wip` file controls how the user settings are created when the program is first run. For more information, see Customizing the installation with the winteg.wip file, on page 49.

Upgrading the server from an earlier version

When you are upgrading from an earlier version, the user session settings and other personal files are stored in the `Documents and Settings/All Users` folder. Use the wIntegrate Administrator or Monitor to ensure that all user sessions are closed before you install. Then, stop the system service by running the Service Control Program or the equivalent Windows utility.

Note: If you have already been using wIntegrate, make a separate copy of any files in the Program Files folder that you have modified for your own use.

The upgrading procedures are the same as described in Installing the wIntegrate Server, on page 11.

Verifying the installation

You can use the wCheck program to verify some aspects of the wIntegrate installation.

About this task

Windows uses its Registry to store references to applications and their components. You can use the wCheck program to view these settings. The wIntegrate installation process normally registers these items automatically. The wCheck program checks that they are registered successfully and enables you to register them if necessary. This is not normally required, but it is a useful troubleshooting aid if you have any unexpected issues with the application.

Note: A full discussion of modifying the Registry is beyond the scope of this document. However, what you should see in the wCheck program is that all of the components are checked to denote that they are registered, they are in the same folder location, and they are of the same version.

If the Registry becomes corrupted for any reason, in addition to using wCheck, you can also refresh the wIntegrate entries by uninstalling and reinstalling wIntegrate.

For information about how to check the installation by using wCheck at the command line, see Verifying the installation with command line parameters, on page 19.

Procedure

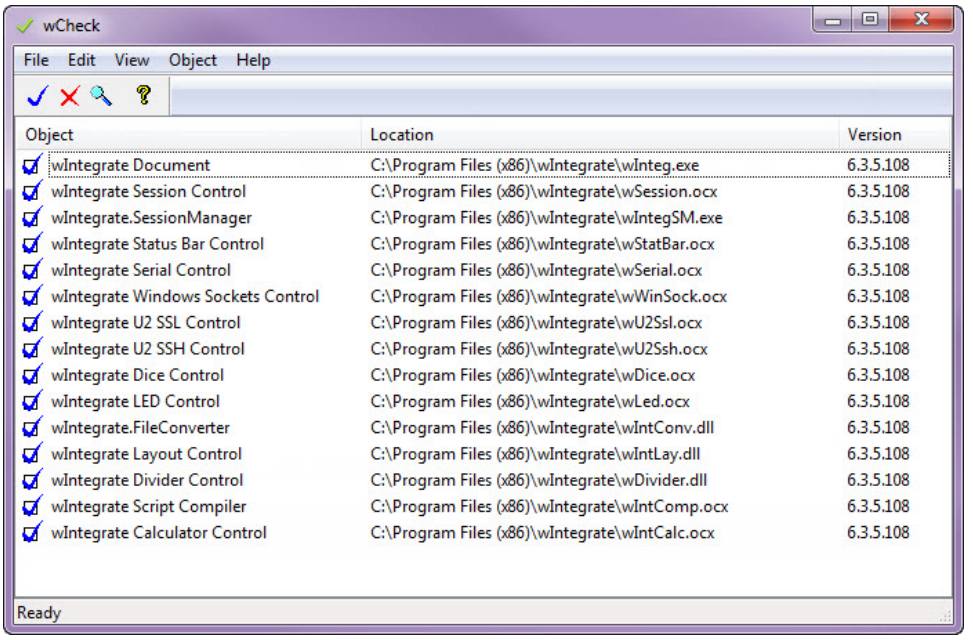
1. Click the Windows **Start** button, and select **All Programs > wIntegrate > Check Installation**.

   To check the installation for the wIntegrate Server, select **All Programs > wIntegrate Server > Check Installation**.

   If you are using Windows Vista or Windows 7, right-click **Check Installation** and select **Run as administrator**.

   The wCheck program appears with all the objects installed listed.

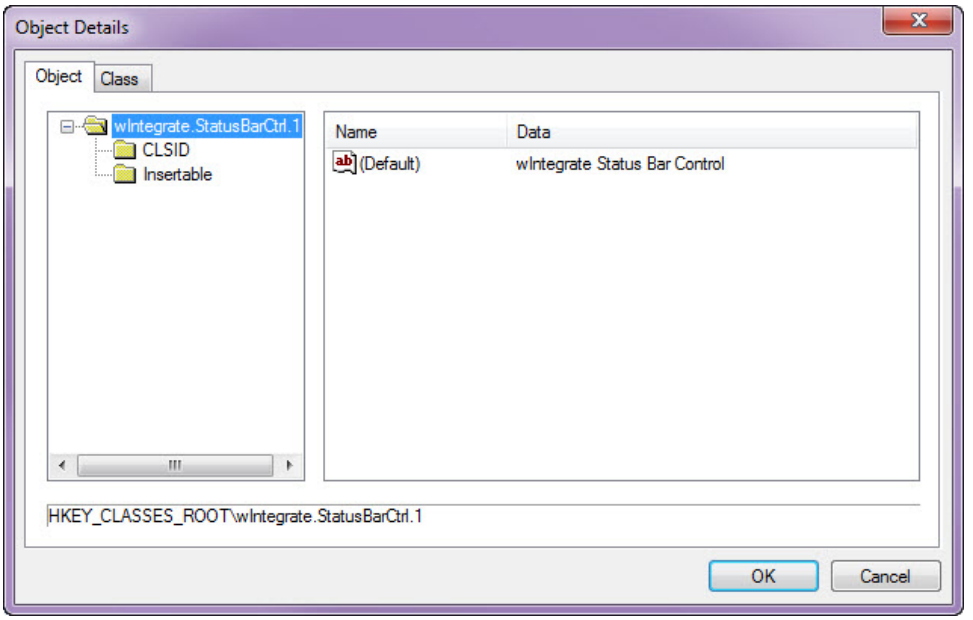
   Figure 3: wCheck



   If the check box for a wIntegrate object is selected, the object is registered. Typically, all objects should be installed to the same folder.

2. Double-click an object to view more information about it.
   The Object Details appear.

   Figure 4: Object Details



3. Click **OK** when you are done viewing the details.

Registering or unregistering objects

If the need arises, you can use the wCheck program to register and unregister wIntegrate and non-wIntegrate objects.

1. To register a wIntegrate object, from the wCheck program window, select a wIntegrate object, and click the **Register File** icon ( ).

   Tip: The list might not update automatically. To manually refresh it, select **View > Refresh** from the toolbar.

2. To register a non-wIntegrate object, from the toolbar, select **File > Register**.
3. In the Register File dialog box, select the object that you want to register, and click **Open**.
4. To unregister an object, select the **Unregister File** icon ( ).
5. To unregister a non-wIntegrate object, from the toolbar, select **File > Unregister**.
6. In the Unregister File dialog box, select the object that you want to unregister, and click **Open**.

Verifying the installation with command line parameters

Most of the functionality of wCheck can be performed by running wCheck with command line options.

Command line options are not case sensitive and can be specified with either a forward slash (/) or a hyphen (-). Additional arguments are shown in parentheses; angle brackets (< and >) specify required arguments and square brackets ([and]) specify optional arguments. The vertical bar (|) denotes that

one or more arguments can be specified. All arguments that contain spaces should be surrounded by double quotation marks ("). All options can be combined.

- Use `RegServer ListOfFiles` to register an object from its file, for example,

  ```
  wCheck -RegServer "C:\Program Files\wIntegrate\wGrid.ocx" "C:\wLed.ocx"
  ```

- Use `UnregServer ListOfFiles` to unregister an object to its given file name, for example,

  ```
  wCheck -UnregServer "C:\Program Files\wIntegrate\wDice.ocx"
  ```

- Use `Register ListOfObjects | FolderName` to register a wIntegrate object given its name, for example,

  ```
  wCheck -Register "wIntegrate Windows Sockets Control"
  wCheck -Register "C:\Backup\wIntegrate"
  wCheck -Register
  ```

  If a folder name is specified, all wIntegrate objects with associated files in that folder will be registered. If no objects are specified, all wIntegrate objects are registered.

- Use `Unregister ListOfObjects | FolderName` to unregister a wIntegrate object given its name, for example,

  ```
  wCheck -Unregister "wIntegrate Serial Control"
  wCheck -Unregister "C:\Backup\wIntegrate"
  wCheck -Unregister
  ```

  If a folder name is specified, all wIntegrate objects with associated files in the folder will be unregistered. If no objects are specified, all wIntegrate objects are unregistered.

- Use `Update` to register all currently unregistered wIntegrate objects, for example,

  ```
  wCheck -Update
  ```

- Use `Report` to display which objects were registered and unregistered by the program, for example,

  ```
  wCheck -Unregister -Report
  ```

- Use `NoErrors` to inhibit the display of any errors during object registration, for example,

  ```
  wCheck -Register -NoErrors
  ```

Modifying a wIntegrate installation

Rerunning the `AutoRun.exe` setup program allows you to modify, repair, or remove an application. The modify option might not be available in cases, depending on the suboptions available.

The installer will not reinstall files that already exist, so if you want to completely refresh the installation for any reason, you have to uninstall then reinstall. This does not affect user settings, which are stored in My Documents.

Uninstalling wIntegrate

If you need to uninstall wIntegrate for any reason, follow these uninstallation steps.

1. Navigate to **Control Panel > All Control Panel Items > Programs and Features**.
2. For a complete uninstallation, individually uninstall all of the following components by selecting one at a time, and then clicking **Uninstall**:
   - wIntegrate
   - wIntegrate Administrator

- wIntegrate Client
- wIntegrate Manuals
- wIntegrate Monitor
- wIntegrate Server

3. If you did not add or modify any of the files to the folders in the `Program Files (x86)` location, the folders will be deleted when you uninstall the programs. However, if you need to delete them manually, they are:
   - `wIntAdm`
   - `wIntCli`
   - `wIntegrate`
   - `wIntMon`
   - `wIntServ`
4. **Optional:** To remove user settings and license information, delete the following folders:
   - `Documents\wIntegrate`
   - `C:\ProgramData\Rocket Software\wIntegrate Licence`

If you initially installed the wIntegrate Server and local client on the same computer, uninstalling the wIntegrate Server might corrupt the local installation. This is because some components are used by both versions and are present in both folders (`Program Files (x86)\wIntegrate` and `Program Files (x86)\wIntServ`). The Registry will normally ensure that the most recently installed version of the component is used, but if you uninstall one product, you might be removing Registry entries used by the other.

The best way to fix corrupt Registry entries is to re-install the program that has stopped working. You can also use the Check Installation program to view and update Registry entries. For more information about the Check Installation program, see Verifying the installation, on page 17.

Installing host programs

The wIntegrate host programs support file transfers and other advanced communication between the PC and the host computer. You must install these programs before you can transfer information and perform other operations between the host computer and the PC. New versions of the host programs are always backward-compatible with earlier releases of the program. New releases do not disable earlier functionality.

Prerequisites

- The wIntegrate communications protocol must match the host computer protocol.

About this task

Because some wIntegrate host programs are host-dependent, you must specify the type of host computer that you are using.

The host programs consist of the file transfer and host API programs. The two types of programs can be installed at the same time or separately. If you want to use wIntegrate's Query Builder to create reports, you must install the host API.

You can install the host programs in any account on your system where you have adequate permissions to create files and compile and catalog programs.

Note: If the host programs are already installed in another account on the host computer, you can share the programs by using the WIN.SHARE program. See Sharing the host programs, on page 23.

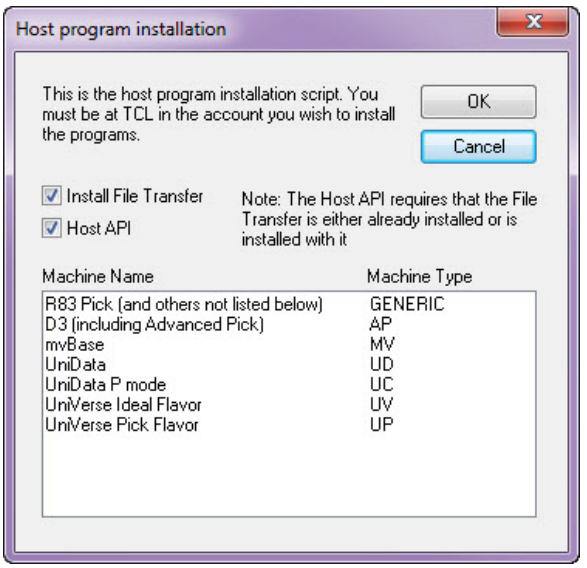
Procedure

1. Start a wIntegrate session, and log on to the host computer.
2. Open the host system database prompt.
3. From the wIntegrate window toolbar, select **Run > Script**.

   A Run Script file browser appears.
4. In the Run Script file browser, navigate to `C:\Program Files (x86)\wIntegrate\Host`, and select **inst_pgm.wis**. Click **Open**.

   The Host program information dialog box appears.

   Figure 5: Host program installation



5. From the Host program installation dialog box, select the check box next to the type of programs you want to install.

   Note: You can install the host API only if you previously installed the file transfer programs or if you are installing the file transfer programs at the same time.

6. In the Machine Name list, select the host database that you are using with wIntegrate.

   If your machine is not listed, select one with a GENERIC type.
7. Click **OK**.

   The installation script performs the following tasks:

   - Creates the WIN.PROGS program directory in the database account.
   - Creates the MACHINE.TYPE record in the WIN.PROGS directory, defining the database/machine type in attribute 1.
   - Installs the WIN.BOOT program in the WIN.PROGS directory.
   - Compiles and runs WIN.BOOT. This program has only one function—to initiate the error-checked export of programs from the PC to the host. wIntegrate exports the same programs to

the host computer regardless of machine type. During export to the host computer, the Export File Monitor appears.
If you chose to install the host API, the file `BP_HAPI.TXT` is also exported to WIN.PROGS.

- Compiles and catalogs the host programs.

8. Verify wIntegrate window reads, “Host Application Program Interface INSTALLED.”

Sharing the host programs

After you install the host programs in one account, you can run the WIN.SHARE program to enable another account to run the host programs

1. Start a wIntegrate session, and log on to the account in which you installed the host programs.
2. Type `WIN.SHARE`, and press **Enter**.
3. Enter the full UNIX path for the account to share the host programs, and press **Enter**.

   A pointer is created from the shared account to the host programs account. You can now use the host programs to transfer files into the other account.

Chapter 3: Configuration overview

The following sections provide information and steps to help you configure wIntegrate.

Two configuration files are provided when you first install wIntegrate: `Session1.wic` and `Session2.wic`. All of the session settings—from communications, to colors, to emulation—are stored in a configuration file. If you open more than one wIntegrate session, each session will have its own configuration file.

Configuring the wIntegrate local client

To configure the local client, complete the tasks in this section.

Starting a session

When you start wIntegrate for the first time, the Session Wizard guides you through the process of creating a session.

1. Click the Windows **Start** button and select **All Programs > wIntegrate > wIntegrate**.
2. Follow the steps in the Session Wizard. Click **Finish** to complete the Session Wizard.

   The login for your host computer appears.

After you save the wIntegrate session, the Session Wizard no longer appears each time that you start the application. If you need to use the wizard later, select **File > Session Wizard** from the wIntegrate toolbar.

The wIntegrate sessions are saved as `.wic` (configuration) files in the `My Documents\wIntegrate\Session` folder. You cannot edit these files with a text editor because the files are in a binary format; however, you can use the wIntegrate Editor to edit these files. For more information about the Editor, see the *Client Scripting Reference* guide.

Opening multiple sessions

You can open multiple sessions to the same host or to different hosts, subject to the hosts' ability to support multiple connections.

About this task

When you start the wIntegrate application `wInteg.exe`, wIntegrate opens with the sessions listed in the **Start Up Sessions** tab from **Setup > Application**.

You can also start a specific session `.wic` file by double-clicking it. Many users leave the session `.wic` files in the default `My Documents\wIntegrate\Sessions` folder and put a shortcut to them on the desktop.

wIntegrate uses configuration files with `.wic` extension by default. These `.wic` files are non-editable binary files. You cannot use the same `.wic` file to open many sessions; instead, wIntegrate can make a temporary copy for each additional session. When you use a copy of a `.wic` file, you might want to make it a default session, but it is preferable to save the copied session with a new name by using **File > Save As** and changing the name. When you exit wIntegrate, you can set the session as default.

The number of sessions that can be opened is limited by Windows memory and resources.

Tip: The session `.WIC` files are stored in a binary format, so they cannot be edited by a text editor such as Notepad. However, you can edit them with syntax coloring with the wIntegrate Editor, which is installed with the local version when you enable Developer Additions during installation.

Procedure

1. If you have a session open, and want to open an additional one, from the wIntegrate toolbar, select **File > Another**.
2. From the Open Another Session dialog box, select the `.wic` file for the session that you want to open, and click **Open**.

Setting up default startup sessions

When you start wIntegrate, a single session is opened using `Session1.wic`. If you open more than one session, when you exit, you can configure wIntegrate to open those additional sessions at startup, as well.

Prerequisites

wIntegrate must be configured to display the Exit dialog box on exit in order to set up default startup sessions. To enable this option, select **Setup > Application**, and then select the **Confirm exit** check box.

Procedure

1. To set up a default startup session, from an opened session in wIntegrate, select **File > Exit**.

   The Exit dialog box appears.
2. **Optional:** Select the **Save changes to session parameters** check box.

   All of the settings for the current session such as font size, window size, position, and more options from the Setup > Preferences dialog box are saved to the session file that you are using.
3. Select the **Default startup session** check box.

   The current session starts automatically when you restart wIntegrate. You can select this option for each session that you want to start automatically.

To discontinue the automatic startup of any session, clear the **Default startup session** on the Exit dialog box when exiting the session. Do not delete the `.wic` file without first removing the session from the default startup list.

You can add or remove session from the startup list by selecting **Setup > Application** and clicking the **Start up Sessions** tab.

Setting up shortcuts on the desktop

You might want to set up several wIntegrate shortcuts on your desktop if you run sessions on different host computers, or if you run sessions with different configurations connected to the same host. For example, you might want one session for email and another for a host application.

Prerequisites

Before you start this procedure, open a wIntegrate session and configure it as you want. Save the session as a `.wic` file.

Procedure

1. On your desktop, right-click and select **New > Shortcut**.
2. In the Create Shortcut wizard, navigate to the location of the wIntegrate configuration file for the session for which you want to create a shortcut. Click **Next**.
3. Enter a name for the shortcut. Click **Finish**.
4. To customize the icon that is used for the shortcut, right-click the shortcut and select **Properties**.
5. Click the **Shortcut** tab. Click the **Change Icon** button.
6. Change the path to `C:\Program Files (x86)\wIntegrate\wInteg.exe`, and select a shortcut icon. Click **OK**.

Setting up the printer

You can select a printer for a wIntegrate session without changing the Windows default printer, for example if you want to print on custom stationery as part of your application.

1. From the main toolbar, select **File > Printer Setup**.
2. In the Printer Setup dialog box, select the **Specific Printer** option.
3. Select a printer from the list.
4. **Optional:** Select the other options for font, character mapping, and more as needed.

   If you want wIntegrate to send data directly to the printer without any formatting, select the **Print direct** check box. Use this option to print host data in the same format as you received it.
5. Click **OK**.
6. **Optional:** Save the custom settings as part of the session configuration file by selecting **File > Save** from the toolbar, or select **Save As** if you want to save the session as a new `.wic` file.

Configuring the wIntegrate Server

The wIntegrate Server acts as the middle tier between the Windows or Java the client and the Host. It contains most of the application's functionality. The thin clients essentially act as screen-and-keyboard interfaces to the functionality in the wIntegrate Server.

Managing connectivity

wIntegrate can reconnect sessions that have been disconnected because of a communications break between the Java or Windows thin client and the wIntegrate Server.

When recoverability is enabled and the thin client loses the connection to the wIntegrate Server, the wIntegrate Server keeps the session to the database server open and attempts to reconnect to the thin client.

A dialog box appears on the client, informing the user that the connection to the wIntegrate Server has been lost. It reports the attempts to reconnect to the same session. In the wIntegrate Administrator, you can specify the frequency of attempted reconnections and other parameters.

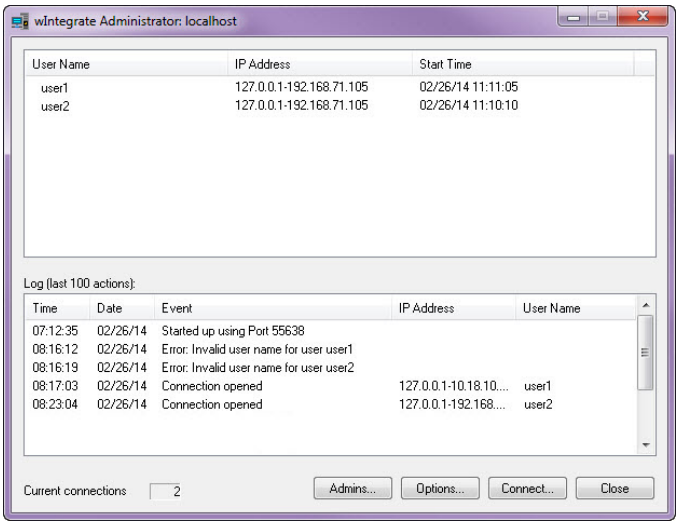
When the reconnection attempts expire, or if the user chooses to stop the reconnection attempts, then the thin client application closes. However, the U2 session on the host is still active and connected to the wIntegrate Server. The administrator might use the wIntegrate Monitor to gracefully end the session.

There is a small performance overhead when using recoverability because of the increased number of messages passing between the thin client and wIntegrate Server.

wIntegrate Administrator

The wIntegrate Administrator is the interface to the wIntegrate Server, which runs as a service and has no interface of its own. The Administrator is a separate application that communicates with the wIntegrate Server. You can run the Administrator connected to the wIntegrate Server running on the same computer or a different computer, and can open multiple instances of the Administrator in case you need to manage wIntegrate servers on multiple machines.

Figure 6: Administrator



The Administrator lists the current connections to the server with the IP address and time and date connected. The log keeps the last 100 actions such as client connections and disconnections to and from the server.

- To open the Administrator, click the Windows **Start** button and select **All Programs > wIntegrate Server > wIntegrate Administrator**
- To connect to the server for which you need to administer, click **Connect**, and enter the user name, password, and server to which you want to connect. Click **OK**.

Using the Administrator to change options for the wIntegrate server

The wIntegrate Server runs as a service. This means it has no interface of its own. The Administrator acts as the interface to the wIntegrate Server, summarizing activity. This is also where options such as recoverability and logging are set.

The settings you see in the Administrator are stored in the Registry in `HKEY_LOCAL_MACHINE \SOFTWARE\Wow6432Node\Rocket Software\wIntegrate Server`.

Note: When options are set in the Administrator, they take effect immediately but they are not saved in the Registry until the wIntegrate Server is closed.

1. From the Administrator, click **Options**.
2. Select the check boxes for the options you want, and clear any that you do not want. Click the different tabs to determine administration options for each category.
   - The **Administrator** tab allows you to change the settings of the Administrator such as connecting at startup and showing local messages. See the following table.

     Table 2: Administrator tab options

Option	Description
Connect at Startup	Select this option for the Administrator to reconnect automatically to the last wIntegrate Server it was connected to.
Show Local Messages	Select this option to show any messages generated locally by the Administrator in the log. Messages from the Server are shown.
Display Exit Confirmation	Select this option to confirm you want to Exit from the Administrator when you click the **Close** button.
SSL Tracing	Select this option to create a trace file for the connection between the wIntegrate Server and the Administrator running on this machine only. **Caution:** This option is retained when you close and restart the Administrator. Use SSL tracing with extreme caution as the trace files can grow until they fill the hard disk.
SSL Trace Folder	Select the folder to store the SSL Trace logs in. The default location is `C:\Users\<username>\Documents`. Log files are keyed by date as `WintAdmin_SSL_`*`YYMMDD_sssss.log`* where *YYMMDD* is today's date, and *sssss* is the number of seconds since the start of the day.

   - The **Server** tab allows you to change options for the wIntegrate Server, as described in the following table.

     Table 3: Server tab options

Option	Description
Log to File	Select this option to log user connectivity actions, as displayed in the main Administrator window. Log files are stored on the wIntegrate Server machine in the `C:\ProgramData\Rocket Software\wIntegrate Server\Logs` folder. Log files are keyed by date as *`YYYYMMDD`*`.log` so a new log file is written every day.
Delete Log Files After *n* Days	Select this option to delete log files after the required number of days.
Client Debug Logging	This option is of limited use and should be avoided unless otherwise advised by Rocket Support. Selecting this option results in large log files for every client. Select this option to log client activity. Log files are written to the `C:\ProgramData\Rocket Software\wIntegrate Server\Logs` folder. Log files are keyed by date, time, user name, and IP address. Only internal activity IDs are written to the file, which might be helpful to Rocket Support.

Option	Description
Enable Remote Administration	Select this option to allow remote users to run the Administrator. Note that the Administrator is always enabled on the machine where the wIntegrate Server is running as it is the only interface to the wIntegrate Server.
Enable Monitor	Select this option to allow the Monitor to be used. This also applies to the machine where the wIntegrate Server is running. Use of the Monitor is subject to user credentials.
Enable Keystrokes in the Monitor	Select this option to allow an authorized user of the Monitor to enter keystrokes in a user session. This is useful to gracefully close sessions for disconnected clients.
SSL	Select this option to encrypt communications between the wIntegrate Server and the Windows/Java thin clients with SSL. If you select this option, you must specify an SSL certificate. SSL options in this dialog are unavailable (grayed out) unless the Administrator is running on the same machine as the wIntegrate Server to which it is connected. If this option is enabled, the Windows and Java thin clients Administrator and Monitor negotiate with the wIntegrate Server regarding which SSL protocol to use. Only TLSv1.1 and TLSv1.2 protocols are used, as earlier protocols are no longer considered secure. The TLSv1.2 protocol is preferred. If you are using the Java client, the Java Control Panel might allow enabling and disabling certain protocols.
SSL Certificate	Use this option to specify which certificate to use as the Server Certificate. The name specified here will be a certificate Common Name. The Select Certificate dialog lets you browse for and select a certificate from one of two certificate stores, Root or CA. These are each in one of two locations, Machine Stores or User Stores. The Root certificate store is sometimes referred to as Trusted Root Certification Authorities and the CA certificate store as Intermediate Certification Authorities.
Disconnection Timeout	This option is used when the recoverability option is not enabled on the **Recoverability** tab. If the **Disconnection Timeout** check box is selected, you can specify how long to keep a host session active after a client has disconnected unexpectedly, for example due to a broken WiFi connection. After this timeout period, the wIntegrate Server will close the server connection, possibly leaving host files or items locked. The allowable range of seconds is 30-86,400 (24 hours).

- The **Security** tab allows you to set user credentials using the Windows Authentication, as described in the following table.

Table 4: Security tab options

Option	Description
Use Windows Authentication	Select this option to enable user authentication using Windows Active Directory. This option is disabled unless Windows Authentication is available. If this option is disabled, users are authenticated according to the credentials created in the User Management window, available from the **Start > wIntegrate Server** menu.
Client Group	Select the Windows Group that has the authority to use the thin client.
Administrator Group	Select the Windows Group that has the authority to use the wIntegrate Administrator.

Option	Description
Monitor Group	Select the Windows Group that has the authority to use the wIntegrate Monitor. Note that the Monitor can be universally disabled in the Server tab.
Monitor Send Group	Select the Windows Group that has the authority to send keystrokes to user applications in the wIntegrate Monitor.
Monitor End Group	Select the Windows Group that has the authority to use the End Session button in the Monitor to end user sessions

- The **Recoverability** tab sets the options for reconnecting thin clients when, for example, they lose the connection due to a poor Wi-Fi signal. Both the wIntegrate Server and the thin clients store messages that are sent to the other in case they must be retransmitted so the other side can pick up from where the messages were lost. The receiving side periodically acknowledges that messages have been received so the sending side can remove them from memory. See the following table for more information.

  Table 5: Recoverability tab options

Option	Description
Enable Recovery	Select this option to enable clients to recover lost connections. If this setting is disabled and the connection between the client and the wIntegrate Server is lost, a client is not able to reconnect to the wIntegrate Server and the connection to the multi-value server is closed. If this setting is enabled and the client loses the connection, the wIntegrate Server maintains an active connection to the multi-value server, and the session can be accessed using the Monitor.
Acknowledge Messages	This setting determines how many messages are stored by the sending computer before the acknowledgment is sent by the receiver.
Message Queue Size	Use this option to set the maximum number of messages that are retained in the sending computers memory.
Time Between Retries	When the connection to the client is lost, this setting determines the time interval between the server trying to reconnect to the client.
Heartbeat	This enables the wIntegrate Server to know the difference between an inactive client and a lost connection. You can use the Heartbeat to send a meaningless message to the client which will respond indicating it is still connected.
Auto End Session	Use this setting to automatically end the session if the client does not reconnect.

- The **Client** tab allows you to set the inactivity timeout of the client. After this period the user will have to login again.
- The **Ports** tab allows you to change the TCP/IP ports for the client, Administrator, or Monitor. These settings must be matched in the respective applications.

  Note: The Administrator port must be 55639. Any other value causes the connection to fail as it is a fixed value in the wIntegrate Server.

3. Click **OK** to save the settings.

Viewing other administrators

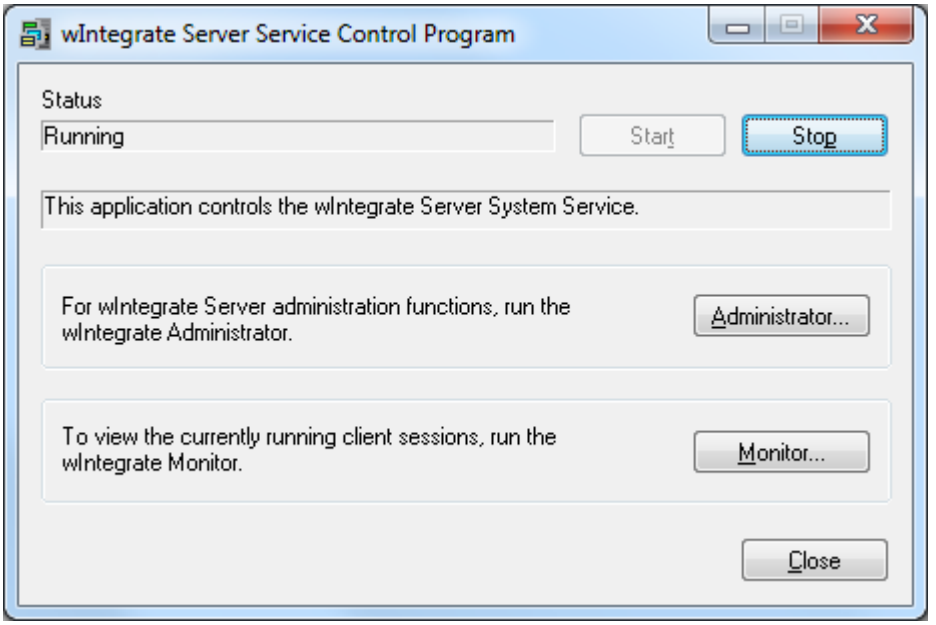
In the Administrator, you can view other users who have Administrator permissions, as set from the User Management program.

From the Administrator, click **Admins**.

A dialog box appears with the active administrators for the server.

Service Control Program

The wIntegrate Server runs as a service and has no interface of its own. The Service Control Program can be used to start and stop the service in the same way that you would from the Windows application that lets you control services.



One reason to stop and restart the service is to, for example, activate a newly installed license with an increased number of users.

To open the Service Control Program, click the Windows **Start** button and select `All Programs > wIntegrate Server > wIntegrate Service Control Program`. Note that the Service Control Program must be started with **Run as administrator**, otherwise it may not have the necessary privileges to access system services.

1. The dialog shows the current status of the wIntegrate Server: **Running** or **Stopped**.
2. If the wIntegrate Server is Stopped, you can click the **Start** button to start it.
3. If the wIntegrate Server is Started, you can click the **Stop** button to stop it.

> **Caution:** Stopping the wIntegrate Server disconnects any users and can leave host files locked. Use the Administrator or Monitor to ensure that no users are connected before stopping the service.

4. Click the **Administrator** button to run the wIntegrate Administrator, which shows the current users.
5. If the Monitor is installed, click the **Monitor** button to run the wIntegrate Monitor. If your privileges allow it, you can connect to any disconnected user sessions and gracefully log off.

Managing users

The wIntegrate Server is supplied with applications to assist with administration.

Four programs are included with the Server installation:

- wIntegrate Administrator
- wIntegrate Monitor (installed separately)
- User Management
- UserStat

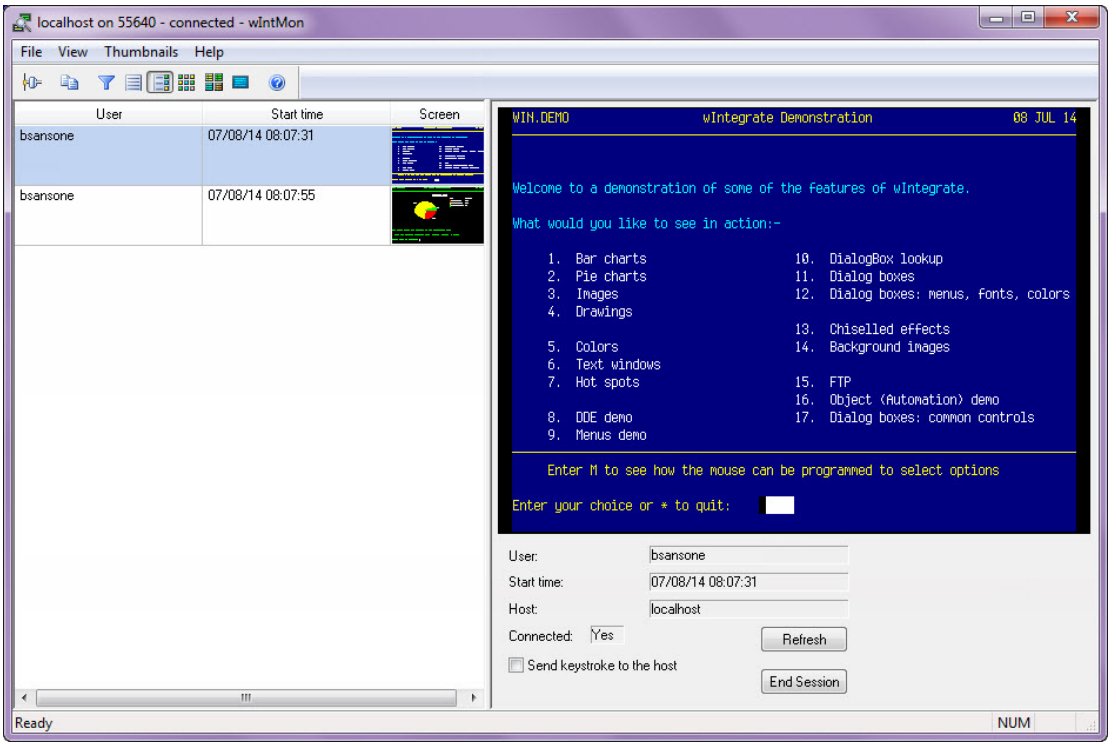
Note: UserStat is installed if Network Administrator Tools was selected during installation. It does not appear in the wIntegrate Server menu. Run the `UserStat.exe` program from the `Program Files (x86)\wIntserv` folder.

The following sections describe how to use these programs.

wIntegrate Monitor

The Monitor lets you see user activity in real time. Any number of simultaneous users can run the Monitor and connect to the same wIntegrate Server. A single computer can run multiple instances of the Monitor, so if you are running the wIntegrate Server on several servers, you can monitor them all from a single desktop.

Figure 7: Monitor



The Monitor shows current client connections with the IP address and time and date connected. You can see character-based user screens in real-time, and even enter data into user screens for support and administration.

- To open the Monitor, click the Windows **Start** button and select **All Programs > wIntegrate Server > wIntegrate Administrator**
  A dialog box appears to connect to the server. You can also connect to the server by clicking **File > Connect**.

Using the Monitor

The Monitor automatically shows all user screens updated in real time as the user sees them. It does not show any dialog boxes that are seen by the user.

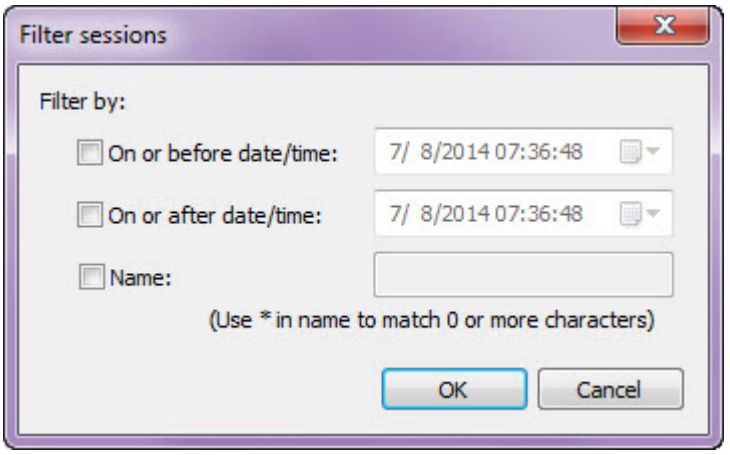
- When you start the Monitor, you must enter your user name, password, and the wIntegrate Server that you want to connect to. The TCP/IP Port must be 55640. Any other value causes the connection to fail as it is a fixed value in the wIntegrate Server.
- Click the **Filter** icon to filter the displayed sessions. For more information, see Filtering and sorting sessions, on page 33.
- Click the icons to change how the thumbnails are displayed. You can view text only, view as a list, and more. You can also change the display from the **View** menu.
- Click the column headings to sort by user or start time.
- Click the **Copy user list to Clipboard** icon to copy to the clipboard the sessions that are listed in the Monitor. Once copied, you can paste them in a different application such as Excel.
- Select the **Send Keystrokes to Host** check box to enter data into the user screen. This option is useful for logging off remote sessions for backups. The user is able to see everything you enter. With the check box selected, you must also click the user screen before characters can be entered. As a warning, a red border appears around the screen when entered characters go to the user session. This check box is selected only if the user of the wIntegrate Monitor has permission to use it, as specified in either the User Management program, or on the **Security** tab of the Administrator if Windows Authentication is used.
- Click the **End Session** button to end a user's session without warning.

Filtering and sorting sessions

The Monitor allows you to filter and sort the connections that are being viewed. Wildcards (*) allow you to filter user names, and you can also specify login date/times to sort by.

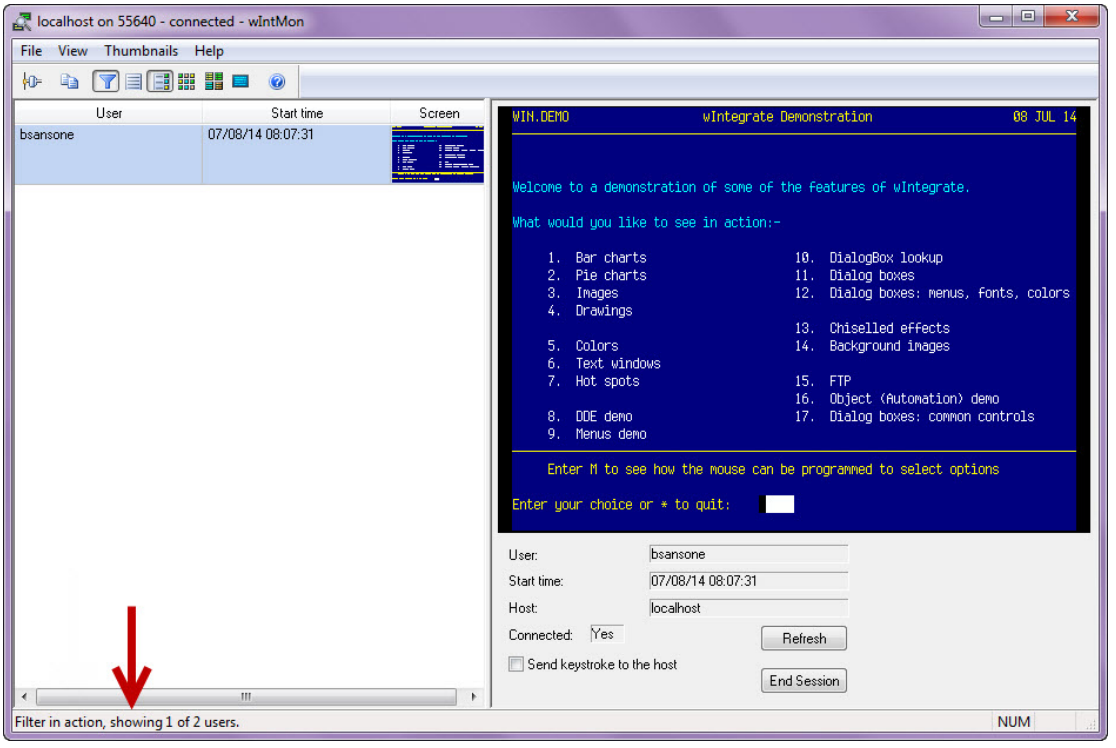
1. From the Monitor, click the **Filter** icon ( ), or select **View > Filter**.
   The Filter sessions dialog box appears.

   Figure 8: Filter sessions



2. Select the check box next to the filtering option that you want to apply. If you select **Name**, you can use the * wildcard to use as a filter at the beginning, middle, or end of a user name. When you are done selecting your filter, click **OK**.
   When filters are applied, a message is displayed at the bottom of the Monitor, as shown in the following figure.

   Figure 9: Monitor with filter

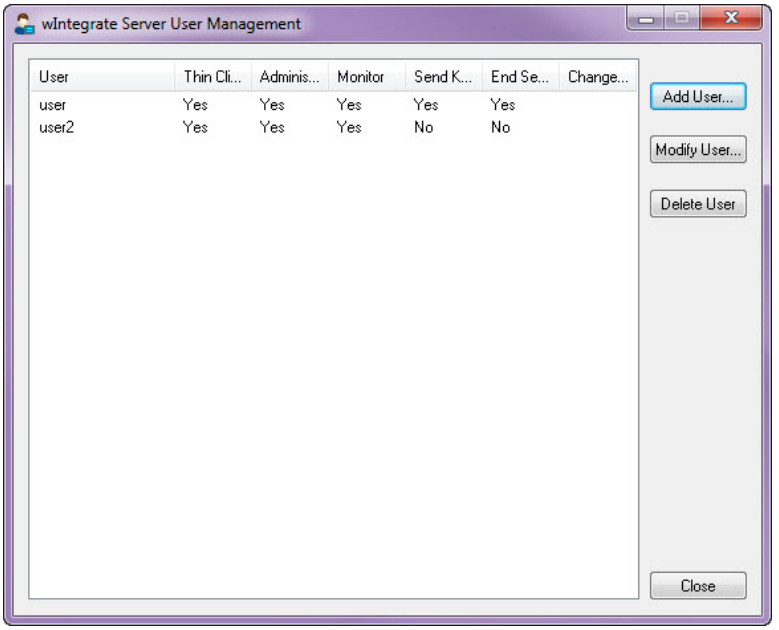


3. To remove a filter, click the **Filter** icon, clear the check boxes, and click **OK**.

User Management

The User Management program allows you to view, add, or delete all of the users, and enable or disable features to a given user. You should set up users in this program if you do not use Active Directory for user authentication.

Figure 10: User Management



- To open User Management, click the Windows **Start** button and select **All Programs > wIntegrate Server > User Management**.

Adding a user

Thin client users must log in when they connect to the wIntegrate Server. If you are not using Active Directory, you must set up users manually in the User Management program and set their permissions using the following procedure.

1. From the User Management program, click **Add User**. Enter the new user's information, and select the check boxes for the programs you want the user to be able to use. You can enable use for the thin client, Administrator, and Monitor. Click **OK** to save the user and settings.
2. If you want to modify an existing user, for example, to enable more privileges, click **Modify Users**.

Backing up server files

The user-specific files for the wIntegrate Server are stored on the machine where the wIntegrate Server runs. The default folder is `C:\ProgramData\Rocket Software\wIntegrate Server\Users\`*`myname`* where *myname* is your user name that is entered when logging on to the server.

With the wIntegrate Server, the user's application-level settings are not stored in the Registry. They are stored in the `Default.wip` file in the `C:\ProgramData\Rocket Software\wIntegrate Server\Users\`*`myname`*`\Profile` location.

To back up the wIntegrate Server's user files, copy the `C:\ProgramData\Rocket Software\wIntegrate Server` folder to a different secure location.

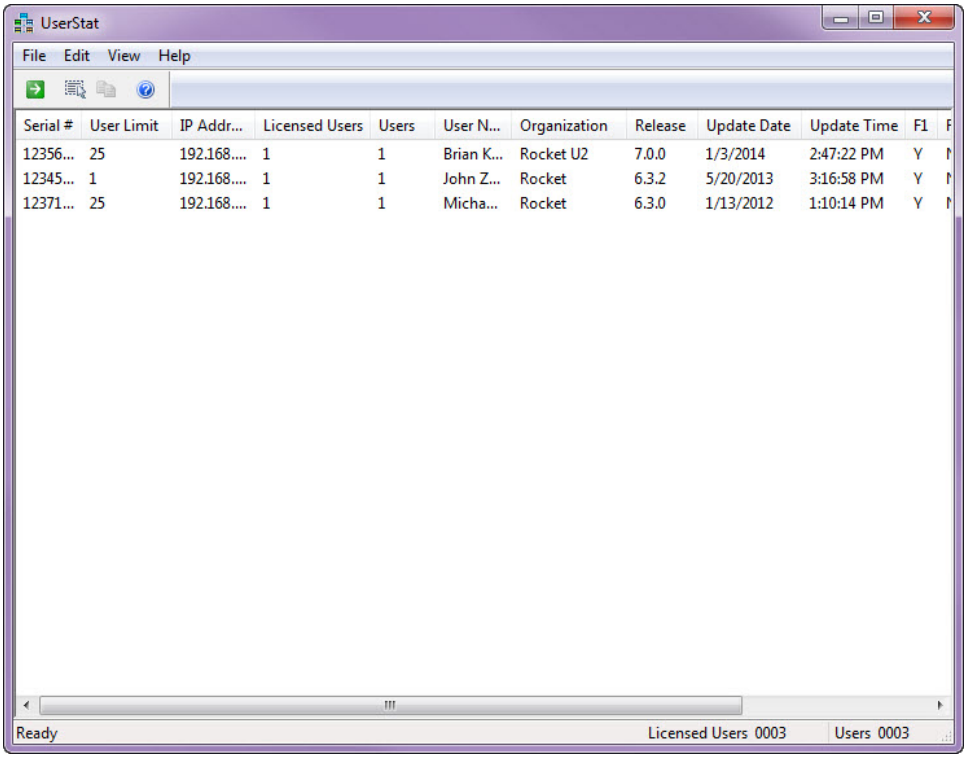
Settings for the wIntegrate Server (as seen in the Administrator and Monitor) are stored in the Registry, in `HKEY_LOCAL_MACHINE\SOFTWARE\Rocket Software\wIntegrate Server`.

UserStat

UserStat shows all wIntegrate users currently running on the network and their license attributes. It shows users of both the local version and the wIntegrate Server. It is invaluable for administrators who are wanting to see what serial numbers are in use.

Attention: Do not run UserStat at the same time as wIntegrate because only one application can use the broadcast port at a time.

Figure 11: UserStat



- UserStat is not installed to the wIntegrate Server menu. To open UserStat, navigate to `C:\Program Files (x86)\wIntServ`, and open `UserStat.exe`.

Viewing all current users

You can see information about all of the current users who are running on the network.

From the UserStat program, select **File > Check**, or click the **Check** icon ( ).

Configuring the Windows thin client

When you start the wIntegrate thin client for the first time, the Session Wizard guides you through the process of creating a session.

Starting a session

1. Click the Windows **Start** button and select **All Programs** > **wIntegrate** > **wIntegrate**.
2. Follow the steps in the Session Wizard. Click **Finish** to complete the Session Wizard.

   The login for your host computer appears.

After you save the wIntegrate session, the Session Wizard no longer appears each time that you start the application. If you need to use the wizard later, select **File** > **Session Wizard** from the wIntegrate toolbar.

The wIntegrate sessions are saved as `.wic` (configuration) files on the wIntegrate Server machine in the `C:\ProgramData\Rocket Software\wIntegrate Server\Users\<user name>\Session` folder. This is the location for all the settings that are stored in My Documents on the local version.

If this folder does not exist, it is automatically created when, for example, an authorized user logs in for the first time. The user's settings folder (known in scripting as SettingsDir) is automatically populated according to the values in the `winteg.wip` (parameters) file. See Appendix C for information on customizing this file to change how these settings are created.

The benefit of storing user files centrally is that the user can log in from any device and the same settings will be used.

You cannot edit the session `.wic` files with a text editor because the files are in a binary format; however, you can use the wIntegrate Editor to edit these files. For more information about the Editor, see the *Client Scripting Reference guide*.

Opening multiple sessions

You can open multiple sessions to the same host or to different hosts, subject to the hosts' ability to support multiple connections. Note that each session must use a different session `.wic` file.

About this task

When you start the wIntegrate application wInteg.exe, wIntegrate opens with the sessions listed in the Start Up Sessions tab from **Setup** > **Application**.

You can also start a specific session `.wic` file by double-clicking it. Many users leave the session `.wic` files in the default `My Documents\wIntegrate\Sessions` folder and put a shortcut to them on the desktop.

wIntegrate uses configuration files with `.wic` extension by default. You cannot use the same `.wic` file to open multiple sessions; instead, wIntegrate makes a temporary copy for each additional session. When you use a copy of a `.wic` file, you might want to make it a default session, but it is preferable to save the copied session with a new name using **File** > **Save As** and changing the name. When you exit wIntegrate, you can set the session as the default.

Tip: The session `.WIC` files are stored in a binary format, so they cannot be edited by a text editor such as Notepad. However, you can edit them with syntax coloring using the wIntegrate Editor, which is installed with the local version when you enable Developer Additions during installation.

Procedure

1. If you have a session open, and want to open an additional one, from the wIntegrate toolbar, select **File** > **Another**.

2. From the Open Another Session dialog box, select the `.wic` file for the session that you want to open and click **Open**.

Setting up default startup sessions

When you start wIntegrate, a single session is opened using `Session1.wic`. If you open more than one session, when you exit, you can configure wIntegrate to open those additional sessions at startup.

Prerequisite

wIntegrate must be configured to display the Exit dialog box on exit in order to set up default startup sessions. To enable this option, select **Setup** > **Application**, and then select the **Confirm exit** check box.

Procedure

1. To set up a default startup session, from an open session in wIntegrate, select **File** > **Exit**. The Exit dialog box appears.
2. Select the **Save changes to session parameters** check box. All of the settings for the current session such as font size, window size, position, and more options from the **Setup** > **Preferences** dialog box are saved to the session file that you are using.
3. Select the **Default startup session** check box. The current session starts automatically when you restart wIntegrate. You can select this option for each session that you want to start automatically.

To discontinue the automatic startup of any session, clear the **Default startup session** on the Exit dialog box when exiting the session. Do not delete the `.wic` file without first removing the session from the default startup list.

You can add or remove sessions from the startup list by selecting **Setup** > **Application** and clicking the **Start up Sessions** tab.

Setting up shortcuts on the desktop

It can be helpful to set up several wIntegrate shortcuts on your desktop if you run sessions on different host computers, or if you run sessions with different configurations connected to the same host. For example, you might want one session for email and another for a host application

Prerequisite

Before you start this procedure, open and configure a wIntegrate session. Save the session as a `.wic` file.

Procedure

1. On your desktop, right-click and select **New** > **Shortcut**.
2. In the Create Shortcut wizard, navigate to the location of the wIntegrate `.wic` configuration file for the session for which you want to create a shortcut. Click **Next**.
3. Enter a name for the shortcut. Click **Finish**.
4. To customize the icon that is used for the shortcut, right-click the shortcut and select **Properties**.
5. Click the **Shortcut** tab. Click the **Change Icon** button.

6. Change the path to `C:\Program Files (x86)\wIntegrate\wInteg.exe`, and select a shortcut icon. Click **OK**.

Storing wIntegrate Server login parameters

The desktop shortcut can specify the parameters that the thin client uses when connecting to the wIntegrate Server. This shortcut saves you time because you do not have to retype parameters such as user, password, server, etc. each time you log in.

Example

When running the thin client, assume that you log in to the wIntegrate Server running on a computer called `OurFileServer` and your user name is `Myname`. If you want to connect to a U2 server with the session `mysession.wic`, you would use the following command line:

```
"C:\Program Files (x86)\wIntCli\wIntCli.exe" -user Myname -password p455w0rd -server
      OurFileServer -autologin Y -session Session2.wic
```

Parameter	Description
user	The user name. If not specified, the default value is remembered from the last name or IP address used to log in. The user name determines the folder with the user's settings in: `C:\ProgramData\Rocket Software\wIntegrate Server\Users` on the wIntegrate Server.
password	The password to log in to the wIntegrate Server. If not specified, this value is not remembered from previous logins.
server	The name or IP address of the machine with the wIntegrate Server. If not specified, the default value is remembered from the last name or IP address used to log in.
session	The session `.wic` file to use. If not specified, the default value(s) are the one or more default startup sessions.
autologon	Enter `Y` to suppress the thin client login dialog. For the autologin parameter to take effect, the name, password and server parameters must be specified on the command line, and not remembered from the last use.
tcpipport	The TCP/IP port used to connect to the wIntegrate Server. This value must match the server-side setting configured in the wIntegrate **Administrator, Options, Ports** tab. If not specified, the default value is 55638.

All fields are optional.

Configuring the Java thin client

Storing wIntegrate Server login parameters

You can create a shortcut on the desktop that specifies the parameters that the Java client uses when connecting to the wIntegrate Server. This shortcut saves you time because you do not have to retype parameters such as user, password, server, autologin, session, and style sheet every time you log in.

Example

When running the Java client as an application, assume that you log in to the wIntegrate Server that is running on a computer that is called *OurFileServer* and your user name is *Myname*. You want to connect to a U2 server with the session `mysession.wic`. You would use the following command line:

```
"\\OurFileServer\c\Program Files\wIntServ\wIntJava\wIntJavaApplet.jar"
-user Myname -server OurFileServer -session mysession.wic
```

When running the Java applet in an HTML page, use the supplied `wIntJavaPage2.htm` file, which accepts parameters in the URL. For example:

```
http://www.myorganization.com/wIntServ/wIntJava/wIntJavaScreen.htm?
name=Myname&server=OurFileServer&session=mysession.wic
```

Applet security

If the wIntegrate Server is not on the same computer as the Web Server, then security must be set to allow the applet to connect to another server.

The Java thin client has an embedded SSL code-signing certificate identifying the publisher as Rocket Software. When the applet runs, a dialog box prompts about the certificate. Click **Yes** or **Always**.

The certificate has a limited life; typically three years. When the certificate expires a message displays, and the applet will not run in a browser. However, at the time of writing, the expired certificate does not prevent the Java client from running as an application (i.e., looking similar to the regular Windows version).

It is possible to check the details of the embedded certificate using the jarsigner program. This is installed with the Java SE Development Kit which can be downloaded from the Java site.

```
jarsigner -verify -verbose -certs c:\<pathname>\wIntJavaApplet.jar
```

Note: To avoid issues specifying the path, copy the `wIntJavaApplet.jar` file to the same folder as `jarsigner.exe`, and run the command without specifying the path.

How the Java thin client is used

The Java thin client is supported as an application, in addition to working as an applet in a web page.

When running as an application, the Java client does not run in a browser, but has its own border, menu and toolbars. So it works in the same way as the classic Windows local version. However, it continues to connect to the wIntegrate Server, which in turn connects to the host.

Running the Java client as an application has some advantages over running it in a browser. The browser's menu and toolbars are not present, removing unnecessary options. Space is saved as the emulation screen is not embedded in an HTML page.

The benefit of embedding the Java client in an HTML page is that it can interact with objects on the page, allowing two-way communications between the HTML objects and the multi-valued host.

The Java client application is shown on the **Start** menu with other wIntegrate Server items. You can run it by executing the `wIntJavaApplet.jar` file. For deployment to end-users, you might prefer to use Java Web Start.

Java thin client as an applet in an HTML page

The files for the Java thin client are installed with the wIntegrate Server, provided the Java thin client option is enabled during installation. These files are installed in the `Program Files\wIntServ\wIntJava` folder and include sample HTML pages to help you get started.

Table 6: Java thin client files

File	Description
`wIntJavaPage.htm`	Open this HTML page in your browser to run the wIntegrate Java thin client. There are two ways to invoke the Java thin client from HTML. The recommended technique is to use the <APPLET> tag. This maximizes compatibility across all platforms and browsers. An alternative technique is to use the <OBJECT> tag with the Java plugin ActiveX control. As this technique requires ActiveX support, it works only on Windows platforms. Note that it is the **classid** field that specifies that Java run time is present, otherwise it is downloaded. You might want to edit or rename the `wIntJavaPage.htm` file to change the PARAM fields **host** and **username** as appropriate for your wIntegrate Server. These are the same fields that you would enter into the login for the Windows thin client, and not the login to your host U2 server.
`wIntJavaPage2.htm`	This example shows how to pass parameters from the command line to the Java applet.
`wIntJavaApplet.jar`	This file is the code for the Java thin client. It has to be downloaded to the client, at least for the first execution. Depending on the settings on the client computer, this file is usually cached.
`Instructions.htm`	A simple HTML page describing how to use the Java thin client.
`wIntjc.css`	The style sheet stylesheet that is used by the HTML pages. Style sheets control the appearance of the dialogs. They are powerful and flexible. Different styles are applied to different types of control. Styles can be set globally for all dialogs, or specific dialogs, or specific controls within a single dialog. For an example of style sheets, in the Java thin client bring up the Run Script menu, and run the `WebStyle.wis` script in `Program Files (x86)\wIntServ\Example\Script`.
`wIntJava.gif`	This is an image that is shown on the initial login to the wIntegrate Server. If you want, you can replace this file with your own image, which should also be called `wIntJava.gif`. The standard image is 32x32 pixels in the format of an icon. Your image can be any size. You might want to show different images from different servers
`wIntJavaHelp.zip`	A zipped file containing help in HTML format. If you want to make help available to users of the Java thin client, these files should be copied to your web server. These Help pages are generated from the same source as the Help seen in the Windows versions.

Java thin client as a stand-alone application

When running as a stand-alone application, the Java client looks and feels very similar to the Windows version.

Table 7: Java thin client files

File	Description
`wIntJavaApplet.jar`	This is the only file required. It has to be downloaded to the client, at least for the first execution. Depending on the settings on the client computer, this file is usually cached.

Missing features in the Java client

The Java client is not 100% backward-compatible with the Windows version of wIntegrate. Some of the missing features cannot easily be supported by Java, while others have not been implemented yet because of time constraints.

The following list outlines the current missing features.

- Status bar does not show CAPS, NUM, and other status fields.
- FTP is not supported.
- Some developer features are not fully supported. For example, in the grid, not all of the methods and properties are supported.
- In **File > Printer Setup**, the **Setup** button normally displays the setup dialog for the selected printer. This button is disabled.
- Not all of the options on the right-click **Copy** context menu are supported.
- In **Setup > Character**, the default sessions use Consolas for the character display. The Java client uses Courier instead of the expected Consolas. This is intentional as Consolas might not be available on the client.
- The Report Wizard is not supported because the dialog uses an ActiveX component, which is specific to Windows.
- Enable/disable terminal screen.
- Scripting: Getting font or system information from the client.
- Scripting: Session display functions are not supported, including Session Enable/Locksize/Move/Resize.

On Macintosh, the F9, F10, and F11 keys are used by the system to tile and hide windows. These keys are therefore not available to wIntegrate. If your application uses these keys, consider modifying your software to use, for example, Shift combinations instead

The licensing mechanism

When a wIntegrate client first starts, it broadcasts its serial number over the network. Any other wIntegrate client on the network running version 6.2.0 or later will respond with its own serial number.

The broadcasting client checks these responses and any of the following may happen:

- If it finds no other client with the same serial number, the program continues to load normally.

- If it finds other clients using the same serial number as its own, it checks the key to determine if the user limit has been reached. If the key will allow another client to use the serial number, wIntegrate loads normally.
- If it determines that all available licenses for its serial number are in use, it displays a warning message asking the user to inform the system administrator that insufficient licenses are deployed. The program then closes after allowing time for the user to log out.

The license broadcast is made when wIntegrate starts on the client, and behaves the same for all versions of the program whether running from a stand-alone PC or from a server, or using a single-user or multi-user license. When running wIntegrate as a thin client or Java thin client it is the wIntegrate Server which handles broadcasts and licensing, not the client which simply acts as a screen-and-keyboard interface for the server.

A license is backward-compatible with earlier versions as far as v6.2.0, but not for later versions than the one for which it was created. So if you have a license supplied with v6.3.6 it will also work with v6.2.0 but not for v6.3.7. You can run any combination of wIntegrate versions within the network with the same serial number provided the key supports the releases you are running. However you can only run one version of wIntegrate on a single PC.

Licensing wIntegrate

During the installation procedure, you enter your license information. However, you might need to enter this information later or need to change it if, for example, you need to increase the number of users or upgrade to a new release.

Note: You can mix any combination of versions on your network using a single serial number, provided the license supports the required number of users and versions that you want to run.

1. Close wIntegrate if it is running.
2. To license wIntegrate, from the Windows **Start** menu, select **All Programs > wIntegrate > Licensing**.
3. Enter your user name and organization. Enter the serial number and key that came with your software package. Click **OK**.

Licensing wIntegrate by using command line parameters

The licensing program `wLicense.exe` should normally be run as part of the installation process. It can accept parameters and run in the background without user intervention.

There are two ways to pass parameters to the licensing program. The switches used by msiexec can change the user experience. You can choose to use either.

Passing parameters on the MSI command line

The four parameters required by the licensing program are:

- LicenseUser: user name
- LicenseOrg: organization
- LicenseSerial: serial number
- LicenseKey: the Key

For example:

```
msiexec /i "wIntegrate.msi" LicenseUser="John Smith"
LicenseOrg="Company Name, Inc." LicenseSerial=12345678 LicenseKey=qwer-
asdf-zxcv-tyui-1
```

Passing parameters directly to the licensing program

As an alternative, the licensing program can be run stand-alone or as part of an MSI installation as follows:

Syntax:

```
wlicense.exe -U<username> -O<organization> -S<serial number> -K<key>
```

For example:

```
wlicense.exe -U"John Smith" -O"Company Name, Inc." -S12345678 -Kqwer-
asdf-zxcv-1
```

Creating a silent install with msiexec

msiexec supports a `/qn` switch which runs the installer silently (without any user interaction). However, this switch cannot work as a single-line command with wIntegrate. This is because the licensing dialog is part of the installation interface, which is suppressed by the `/qn` switch. So, the license is not updated.

If you want to use msiexec with a silent installation, you must execute these three statements at the command line; one to install, one to change directory, and one to create the license.

```
msiexec /i "wIntegrate.msi" /qn
cd C:\Program Files (x86)\wIntegrate
wlicense.exe -U"John Smith" -O"Company Name, Inc." -S12345678 -Kqwer-asdf-zxcv-1
```

Updating the license information

wIntegrate's licensing scheme uses a combination of serial number and authorization key. During the initial licensing process you enter the serial number, user name, organization, and authorization key. The resulting `license.wil` file is written to the `C:\ProgramData\Rocket Software \wIntegrate License` folder.

At a later date, you may need to install a new license, for example when you increase the number of users or upgrade to a new release. Select the **Licensing** option from the **Start > All Programs > wIntegrate** menu. After entering the new key, an updated `license.wil` file is written to the wIntegrate license folder.

A new license for a single serial number need be entered only once on a single PC on the network, as it is automatically updated to other PCs during the start-up licensing checks. As each PC starts the program, it receives licenses from other PCs using the same serial number. If it receives a license with a later time or date than its own, it will automatically update itself.

The time and date is stored inside the license file. It does not use the file attributes. Obviously for this system to work properly, it is critical that the system time and date is correct when any new license details are entered. This is the reason for the warning about the system time and date in the licensing dialog.

Appendix A: Preventing users from running file transfers

You can prevent users from running file transfers. This is useful in the case of audits, where the ease of file transfers in wIntegrate is seen as an issue because users can easily download from and upload to the database.

You can prevent file transfers in three ways:

- In **Setup > Preferences**, click the **Scripts** tab, and from the **Menu Bar Script** drop-down list, select **usermenu** and click **OK**. The `usermenu` script is stored in `C:\Program Files (x86)\wIntegrate\wIntSys\Menu`. It is intended to be customized as appropriate for your users. The standard functionality includes:
  - Removing the **Run > Export File** menu option
  - Disabling the **Run > Bridge Copy File** menu option
  - Adding a new **Batch Queries** menu option
- The file transfer menu options can be removed from the menus by editing the `NewSess.wis` file in the `C:\Program Files (x86)\wIntegrate\wIntSys\Script` folder. While this fixes the interface, it is not secure because users might be able to restore these options and/or run the file transfer in another way.
- The Service Subroutine is called by all the host processes that transfer data. You create this subroutine to filter available files or data according to user privileges. As a Basic subroutine, it should be 100% secure from unauthorized changes.
  For more information about the Service Subroutine, see the information about it in the *Host Subroutine Reference*.

Appendix B: Globalization

In addition to European languages, wIntegrate supports multi-byte languages for input and display on the character screen, including Simplified Chinese, Japanese, and Thai.

Setting up the simplified Chinese version

Simplified Chinese is supported in the local version of wIntegrate; it is not supported in the thin client versions. If you have installed the Microsoft GB18030 Support Package and have the Chinese fonts available, you can force the CD image to display in Simplified Chinese with the following command line options:

`path\Autorun.exe 1033\autorun.ini` where *path* is the installation path.

To force the CD image to display Simplified Chinese text, run `path\Autorun.exe 2052\autorun.ini`.

The Chinese CD image menu options are:

- Installation help text
- wIntegrate local client installation

Running the simplified Chinese version

After installation you can switch between English and Simplified Chinese. Changing the language in wIntegrate also changes the language for other wIntegrate applications.

Prerequisites

Before using Simplified Chinese, you must install the appropriate font on the computer; otherwise menus and dialog boxes show characters as squares or question marks. You can check that fonts are available by running the supplied script `Unicode.wis` in `Program Files (x86)\wIntegrate\Example\Script`. It shows Unicode characters for different languages. If a Simplified Chinese font is installed, you can see it by selecting **CJK Unified Ideographs (4E00-(FAF)** in the drop-down menu.

Procedure

1. Open wIntegrate.
2. From the main toolbar, select **Setup > Application**. Click the **Language** tab.
3. From the **Language** drop-down menu, select **Chinese**.
4. Close and reopen wIntegrate to view the changed setting.

Changing the host encoding

You can change how the characters are converted when they are sent between wIntegrate and the host.

1. Open wIntegrate.
2. From the main toolbar, select **Setup > Terminal**.
3. From the **Host Encoding** drop-down menu, select an encoding type, as described in the following table.

Table 8: Host encoding options

Encoding type	Description
Default	wIntegrate uses Latin I on a single-byte language version of Windows, and “Local” on multibyte Windows. This setting works for western languages.
Latin I	No conversion. This is how previous versions of wIntegrate worked. It is usable only for 8-bit character sets (for example, when European languages are used on the PC and host). In these circumstances, the Local option achieves the same results as Latin I; however, Latin I is faster because it does no conversion.
Local	Uses the same settings as the local Windows computer. Use Local for multi-byte languages other than Simplified Chinese, for example Japanese and Korean. The character set must be the same on Windows and the host. This option also works for single-byte character sets, but Latin I is a better option in these circumstances.
GB18030	Use this option with Simplified Chinese. Characters are sent to and received from the host in the GB18030 format that is required for Chinese systems.
Thai	Use this option with Thai.
UTF-8	UTF-8 is an encoding method for Unicode characters, which is the universal character set designed to encode every written character. UTF-8 is the standard encoding method on Linux. Choose UTF-8 encoding when you want to display characters that are not part of the current Windows language setting. For example, use UTF-8 when the local PC is set to English and the U2 host application is Japanese. UniVerse NLS stores data in UTF-8 format.
UVUFT-8	UVUTF-8 is a modified version of UTF-8, more appropriate for users of multi-valued systems. There are some issues when using UTF-8 with UniVerse NLS because of the characters 255 (IM), 254 (AM), 253 (VM), 252 (SVM), and 251 (Text Mark). These have a special function as delimiters within UniVerse. Internally, wIntegrate uses the following Unicode characters to represent the multi-valued delimiters: 255 (IM) - U+F8FF 254 (AM) - U+F8FE (also known as FM) 253 (VM) - U+F8FD 252 (SVM) - U+F8FC 251 (TM) - U+F8FB There is no visual representation for these characters, so they show on the wIntegrate window as squares.

Changing the language in wIntegrate also changes the language for other wIntegrate applications such as the Editor and Dialog Designer.

Setting up the Thai version

Thai is supported in the local version of wIntegrate; it is not supported in the thin client versions. Thai support is limited to the input and display of Thai characters on the emulation screen and dialog boxes. The application has not been localized for Thai – that is the product CD image, installer, application menus, dialog boxes, and messages are still in English.

Thai is not multi-byte as it uses a 256 character font. It differs from Western languages in that certain sequences of keystrokes cause several characters from the font to be combined in a single character cell on the screen.

To set up the Thai version, do the following:

1. Open wIntegrate.
2. From the main toolbar, select **Setup > Terminal**.
3. From the **Host Encoding** drop-down menu, select **Thai**.

If your system has a Thai font, and you enabled Developer Additions during installation, you can see the characters by running the `Unicode.wis` script in `Program Files (x86)\wIntegrate\Example\Script`, then select **Thai** from the drop-down menu. Use the **Run Script** menu option to run the `Unicode.wis` script.

Appendix C: Customizing the installation with the winteg.wip file

Administrators can customize the installation process by editing the `winteg.wip` file in `C:\Program Files (x86)\wIntegrate`. The `winteg.wip` file is in a simple text format that can be changed by administrators to customize settings.

There are two types of settings:

- Application settings, which can be changed by the user in **Setup > Application**. Application settings are stored in the Registry.
- Session settings, which are stored in the `.WIC` session configuration file. The current session name and location can be seen in the **File > Save As** dialog box. These settings can be changed by the user in **Setup > Preferences**. Multiple sessions can run at the same time, and each one has its own settings. The session `.WIC` file is in binary format. It can be edited in the wIntegrate Editor, which is installed if you selected the Developer Additions during installation.

Application settings are not created during installation; they are created for each logged in user when that user first runs wIntegrate. It is the `winteg.wip` file that determines what settings are created at this time. The `winteg.wip` file is in a simple text format that can be changed by developers to customize settings.

Format

The `winteg.wip` file format is:

```
[Sessions]
parameter_1=value_x
parameter_2=value_y
...
```

Example winteg.wip file

The following example of the `winteg.wip` file contains the default sections and settings that show the type of data required for each parameter:

```
; New user setup file
; Version 6.0.1
; Sets up registry and folder when a user first runs the application

[Sessions]
1=Session1.wic

; Uncomment the next line to prevent more than one session
; Maximum=1

[Settings]
ConfirmExit=1
SaveChangesOnExit=0

; Editors left blank or commented out causes the registry
; to be interrogated for their values
; BitmapEditor=
; HtmlEditor=
; TextEditor=
```

```
PreSessionOpenScript="wIntSys\Script\NewSess.wis"
PostSessionOpenScript=

ConnectMessage=Screen

Libraries=1
Library1="File Transfer"

ShowSessionCopyDialog=1
SnapToSize=1


[Folders]
; Comment out UserDir so that the default value is used
; If the full path is omitted its assumed to be relative
; to the users personal folder (typically "My Documents")
; UserDir=

; All other folders are relative to the UserDir if a
; full path is not used
Session=Session
Color=Color
Default=Default
Keyboard=Keyboard
FileTran=FileTran
Import=
Download=
Queries=Queries
QueryImport=
RWTemplate=RwTemplates
RWImport=
PDFTemplate=PDFTemplate

[Folder Descriptions]
UserDir="Base folder for user created files"
Session="Session configurations"
Color="Color Schemes"
Default="Defaults"
Keyboard="Saved Keyboard settings"
FileTran="Saved File transfers"
Import="Default folder for Imported data"
Download="Default folder for downloaded data"
Queries="Saved Query Builder queries"
QueryImport="Default destination for files imported with the Query Builder"
RWTemplate="Report Wizard Templates"
RWImport="Default destination for files imported from the Report Wizard"
PDFTemplate="PDF query and report formatting templates"


[Copy Folders]
Session=Session
Color=Color
Keyboard=Keyboard
Default=wIntSys\Default
FileTran=FileTran
Queries=Query\QuerySav
RWTemplate=RepWiz\Template
PDFTemplate=PDFTemplate
```

Sections of winteg.wip file

The `winteg.wip` file contains five default sections:

- [Sessions] section
- [Settings] section
- [Folders] section
- [Folder Descriptions] section
- [Copy Folders] section

[Sessions] section

This section determines the default sessions as seen in **Setup > Application** on the **Startup Sessions** tab.

The following example shows the [Sessions] section:

```
[Sessions]
1=Session1.wic

; Uncomment the next line to prevent more than one session
; Maximum=1
```

The following table lists the session parameters.

Table 9: [Sessions] section parameters

Parameter	Description
1=Session1.wic	Each line specifying a session file must begin with a sequential number.
Maximum	Determines the maximum number of sessions this user can run simultaneously. This is useful if, for example, you do not use U2 device licensing and you want to limit the number of U2 licenses a user can consume.

[Settings] section

This section determines the general settings that are used.

The following example shows the [Settings] section:

```
[Settings]
ConfirmExit=1
SaveChangesOnExit=0

; Editors left blank or commented out causes the registry
; to be interrogated for their values
; BitmapEditor=
; HtmlEditor=
; TextEditor=

PreSessionOpenScript="wIntSys\Script\NewSess.wis"
PostSessionOpenScript=
```

```
ConnectMessage=Screen

Libraries=1
Library1="File Transfer"

ShowSessionCopyDialog=1
SnapToSize=1
```

The following table lists the setting parameters.

Table 10: [Settings] section parameters

Parameter	Description
ConfirmExit	Displays a confirmation dialog box when the user attempts to close the application. Available values: ▪ **0** – False ▪ **1** – True
SaveChanges	Saves the current session settings when the application closes. Available values: ▪ **0** – False ▪ **1** – True
SnapToSize	Automatically changes the terminal window to fit the font after dragging the border to resize it. Available values: ▪ **0** – False ▪ **1** – True
ShowSessionCopyDialog	Determines whether a warning dialog appears when opening another session with the same name. Available values: ▪ **0** – False ▪ **1** – True
ConnectMessage	Determines where messages about the connection appear Available values: ▪ **Dialog** ▪ **Status Bar** ▪ **Screen**
BitmapEditor	The editor that is used for bitmaps. If blank, then wIntegrate attempts to determine the most appropriate application. Available values: ▪ *Application name* ▪ *File path*

Parameter	Description
HtmlEditor	The editor that is used for HTML files. If blank, then wIntegrate attempts to determine the most appropriate application. Available values: ▪ *Application name* ▪ *File path*
TextEditor	The editor that is used for text files. If blank, then wIntegrate attempts to determine the most appropriate application. Available values: ▪ *Application name* ▪ *File path*
PreSessionOpenScript	Script to run for each new session before the `.wic` configuration file is loaded. Available values: ▪ *Script name* ▪ *File path*
PostSessionOpenScript	Script to run for each new session after the `.wic` configuration file is loaded. Available values: ▪ *Script name* ▪ *File path*
Libraries	The number of additional libraries to be loaded. For example, file transfer is not integral to the application, but added as a separate module. Available value: ▪ *Integer*
LibraryN=	These are the two permitted libraries. It is not possible for additional libraries to be added; however they can be removed if not needed. Available values: ▪ **"File Transfer"** ▪ **"Binary File Transfer"**

[Folders] section

This section determines the folders that are used by specific parts of the application as seen in **Setup > Application** on the **Folders** tab.

To determine the folder settings from within a script, use the `AppFolder` command. See *wIntegrate Client Scripting* for more information.

The following example shows the [Folders] section:

```
[Folders]
; Comment out UserDir so that the default value is used
; If the full path is omitted its assumed to be relative
; to the users personal folder (typically "My Documents")
; UserDir=
```

```
; All other folders are relative to the UserDir if a
; full path is not used
Session=Session
Color=Color
Default=Default
Keyboard=Keyboard
FileTran=FileTran
Import=
Download=
Queries=Queries
QueryImport=
RWTemplate=RwTemplates
RWImport=
PDFTemplate=PDFTemplate
```

The following table lists the folder parameters.

Table 11: [Folders] section parameters

Parameter	Description
UserDir	Determines the main folder. If blank, the folder is `C:\Users\myname\Documents\wIntegrate`.
Session	The folder where session `.wic` files are stored.
Color	The folder where session color settings are stored.
Default	The folder where session default settings stored.
Keyboard	The folder where session keyboard settings are stored.
FileTran	The default folder for session saved file transfer parameters are stored.
Import	The default folder for files that are created by Import File. If blank, the folder is `C:\Users\myname\Documents\wIntegrate`.
Download	The default folder for files that are created by binary file transfer. If blank, the folder is `C:\Users\myname\Documents\wIntegrate`.
Queries	The default folder for saved queries in the Query Builder.
QueryImport	The default folder for files that are created by the Query Builder.
RWTemplate	The folder where Report Wizard templates are stored.
RWImport	The default folder for files that are created by the Report Wizard.
PDFTemplate	The folder where PDF templates are stored, as used in the Query Builder and Report Wizard.
Scheduler	The folder where files used by the Scheduler are stored.

[Folder Descriptions] section

This section determines the description of the corresponding folder parameters, described in the [Folders] section.

The following example shows the [Folder Descriptions] section:

```
[Folder Descriptions]
UserDir="Base folder for user created files"
Session="Session configurations"
Color="Color Schemes"
Default="Defaults"
Keyboard="Saved Keyboard settings"
```

```
FileTran="Saved File transfers"
Import="Default folder for Imported data"
Download="Default folder for downloaded data"
Queries="Saved Query Builder queries"
QueryImport="Default destination for files imported with the Query Builder"
RWTemplate="Report Wizard Templates"
RWImport="Default destination for files imported from the Report Wizard"
PDFTemplate="PDF query and report formatting templates"
```

See Table 11: [Folders] section parameters, on page 54 for information about the folders to determine their descriptions.

[Copy Folders] section

When you run wIntegrate for the first time, some files are copied from `C:\Program Files (x86)\wIntegrate` to `C:\Users\`*`myname`*`\Documents\wIntegrate`. This section determines which files are copied.

The following example shows the [Copy Folders] section:

```
[Copy Folders]
Session=Session
Color=Color
Keyboard=Keyboard
Default=wIntSys\Default
FileTran=FileTran
Queries=Query\QuerySav
RWTemplate=RepWiz\Template
PDFTemplate=PDFTemplate
```

See Table 11: [Folders] section parameters, on page 54 for information about the folders to determine which ones to copy.

Appendix D: Registered components

The following table describes the components that are registered to the system Registry.

These components are able to be unregistered using the Check Installation program or at the command line. For more information, see Verifying the installation, on page 17 and Verifying the installation with command line parameters, on page 19.

Object	Executable	Description
wIntegrate Document	`wInteg.exe`	The main application. Associated in the Registry with `.wic` files.
Session Control	`wSession.ocx`	The component that does most of the work of the application. Can be used inside other applications, such as Visual Basic.
Session Manager	`wIntegSM.exe`	The Session Manager runs as a separate process handling licensing and any communications between sessions on the same computer.
Status Bar Control	`wStatBar.ocx`	Displays the Status bar.
Serial Control	`wSerial.ocx`	Serial communications, rarely used. If not installed, the Setup Communications dialog gracefully ignores its absence.
Windows Sockets Control	`wWinsock.ocx`	Network communications that are used by Telnet. Widely used.
U2 SSL Control	`wU2SSL.ocx`	Secured network communications using SSL. Need not be installed if not required. If not installed, the Setup Communications dialog box ignores its absence.
U2 SSH Control	`wU2SSH.ocx`	Secured network communications using SSH. Need not be installed if not required. If not installed, the Setup Communications dialog box ignores its absence.
Dice Control	`wDice.ocx`	Supplied as a simple developer example of ActiveX client. See the script `AXCTest.wis` in `Program Files (x86)\wIntegrate\Example\Script`.
LED Control	`wLed.ocx`	Supplied as a simple developer example of using ActiveX control in scripts. See the `DebugRrec.wis` and `XmasTree.wis` scripts in `Program Files(x86)\wIntegrate\Example\Script`.
File Converter	`wIntConv.dll`	Used by the Report Wizard that imports HOLD entry print reports to Excel and HTML.
Layout Control	`wIntlay.dll`	Used by scripting to determine the layout of dialogs.
Divider Control	`wDivider.dll`	Used by scripting to for the Divider control, which resizes a specified area of the dialog box as the window is resized.
Script Compiler	`wIntComp.ocx`	The script compiler that is used by the wIntegrate editor to create compiled `.wix` versions of `.wis` scripts.
Calculator Control	`wIntCalc.ocx`	The calculator control is not used by the application. See the `Calc.wis` script in `Program Files\wIntegrate\Example\Script`.